
VIOLENCE, INSTITUTIONS, AND ECONOMIC GROWTH

Evidencia de Panel para Centroamérica, Colombia y República Dominicana

Autor	Pipeline Económico — Auto-generado
Fecha	30 June 2026
Observaciones	179
Países / Unidades	8
Período temporal	2000–2024
Script maestro	run_pipeline.py
Versión pipeline	2.0

1. Resumen Ejecutivo

Este informe presenta los resultados del pipeline econométrico que estima el efecto causal de la violencia (medida por la tasa de homicidios) sobre las instituciones, la inversión extranjera directa y el crecimiento económico en un panel de países de Centroamérica, Colombia y República Dominicana para el período 2000–2024. La estimación principal emplea efectos fijos bidireccionales (país × año) con errores estándar clusterizados a nivel de país.

Número de países (N)	8
Períodos distintos (T)	24
Observaciones totales	179
Panel balanceado	No
Varianza explicada PC1 (inst.)	0.789
Alfa de Cronbach (inst.)	0.863
EQ1: β (violencia $\rightarrow$ inst.)	-0.1404
EQ1: p-valor (CL)	0.3032
EQ2: β (inst. $\rightarrow$ FDI)	0.9000
EQ2: p-valor (CL)	0.0548
EQ3: β (FDI $\rightarrow$ crecimiento)	0.1371
EQ3: p-valor FDI (CL)	0.2126

■ Lectura de resultados

El coeficiente de EQ1 es negativo, indicando que mayor violencia se asocia con instituciones más débiles, aunque el p-valor sugiere que no se rechaza H_0 con los umbrales convencionales bajo este esquema de inferencia. EQ2 muestra que mejores instituciones atraen más FDI ($p \approx 0.055$, marginalmente significativo). El mecanismo completo (violencia $\rightarrow$ inst. $\rightarrow$ FDI $\rightarrow$ crecimiento) requiere interpretación conjunta de los tres modelos y los resultados bootstrap.

2. Información del Dataset

El panel cubre 8 países de América Central, Colombia y República Dominicana. Los datos provienen del Banco Mundial (WDI, WGI) y fuentes nacionales de estadísticas de crimen.

COL	Colombia
CRI	Costa Rica
DOM	Rep. Dominicana
GTM	Guatemala
HND	Honduras
NIC	Nicaragua
PAN	Panamá
SLV	El Salvador

Período temporal: 2000–2024. Número de períodos distintos observados: 24. El panel no es balanceado.

Varianza within-país

La varianza within-país indica cuánta variación temporal existe dentro de cada unidad. Valores altos justifican el uso de efectos fijos.

homicide_rate_log	53.68%
inst_avg	24.98%
inst_pca	24.54%
fdi_percent_gdp	74.44%
gdp_growth	96.20%
gdp_per_capita_log	52.45%

3. Variables Utilizadas

A continuación se listan las variables dependientes, independientes clave y controles utilizados en el pipeline, junto con su transformación aplicada.

homicide_rate_log	Tasa de homicidios por 100K hab.	log(x+1), lag 1 año	Independiente clave (EQ1)
inst_avg / inst_pca	Índice institucional compuesto	Promedio estandarizado / PC1	Mediador (EQ1→EQ2)
rule_of_law	Estado de derecho (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
control_corruption	Control de corrupción (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
political_stability	Estabilidad política (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
fdi_percent_gdp	IED como % del PIB	Sin transformar	Dependiente EQ2 / Indep. EQ3
gdp_growth	Crecimiento real del PIB (%)	Sin transformar	Dependiente EQ3
gdp_per_capita_log	PIB per cápita (USD cte.)	log(x)	Control
inflation	Inflación anual (%)	Sin transformar	Control
exports_percent_gdp	Exportaciones como % del PIB	Sin transformar	Control
population_log	Población total	log(x)	Control
unemployment	Tasa de desempleo (%)	Sin transformar	Control

Índice Institucional — PCA

Se extrajo el primer componente principal (PC1) de las tres dimensiones WGI como medida sintética de calidad institucional. PC1 explica el 0.8% de la varianza. El alfa de Cronbach es 0.863 (KMO = 0.534).

rule_of_law	0.6300
control_corruption	0.5865
political_stability	0.5091

■ Validez del índice institucional

Un alfa de Cronbach > 0.80 indica alta consistencia interna de las tres dimensiones. El primer componente explica 0.8% de la varianza, justificando la reducción dimensional. Todos los loadings son positivos, confirmando que PC1 captura 'calidad institucional general' y no una dimensión específica.

4. Estadísticos Descriptivos

Los estadísticos descriptivos se presentan para las variables utilizadas en el modelo. Las transformaciones logarítmicas reducen la asimetría de las distribuciones originales.

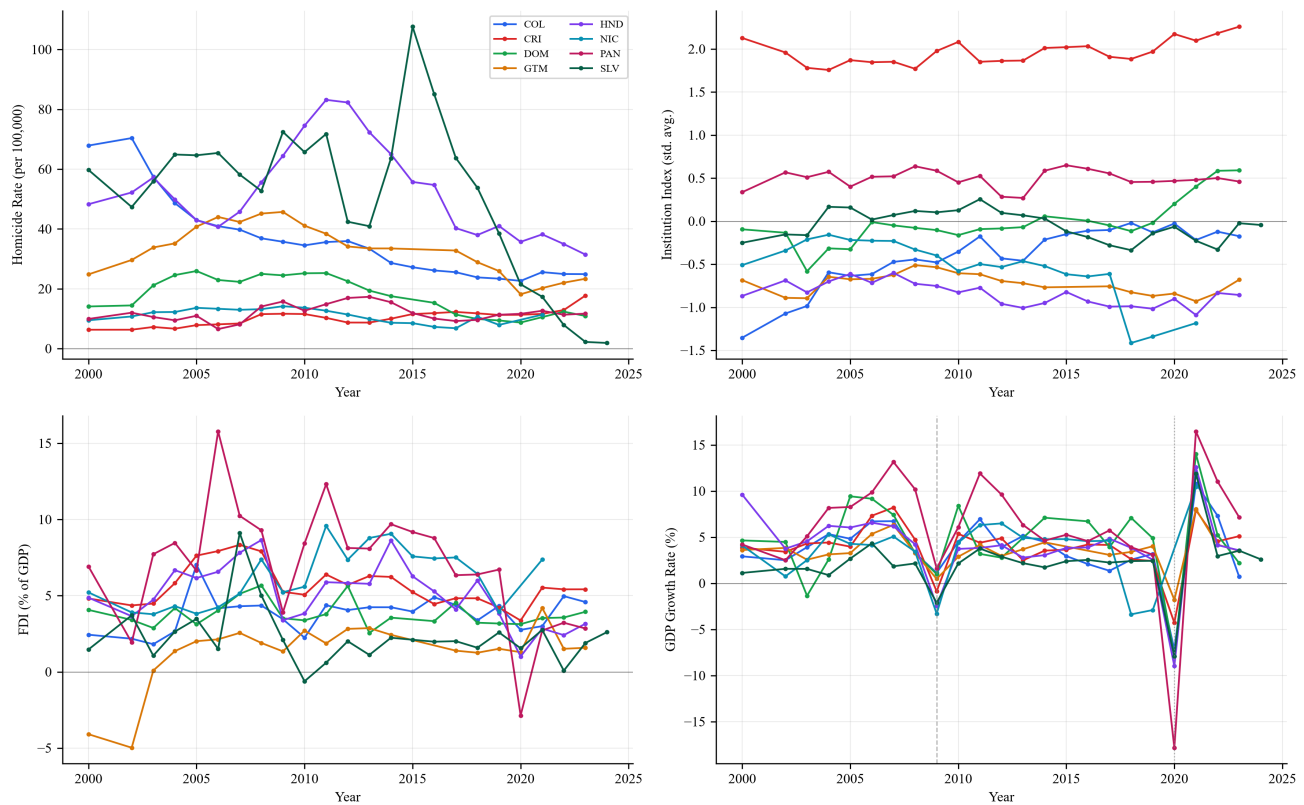
homicide_rate_log	179.0	3.139	0.698	1.86	2.529	3.163	3.734	4.438
fdi_percent_gdp	179.0	4.387	2.722	-4.974	2.654	4.106	5.861	15.779
gdp_growth	179.0	3.975	3.938	-17.821	2.575	3.934	5.29	16.467
gdp_per_capita_log	179.0	8.339	0.706	6.922	7.799	8.299	8.828	9.841
inflation	179.0	5.317	5.058	-1.55	2.676	4.481	6.967	51.461
exports_percent_gdp	179.0	33.998	14.653	13.522	23.95	30.592	42.312	78.299
population_log	179.0	15.988	0.762	14.918	15.439	15.699	16.239	17.773
unemployment	179.0	6.455	3.089	2.143	4.331	5.967	7.716	20.52

Nota: N = número de observaciones no faltantes.

Panel Overview

Figura 1 — Visión general del panel

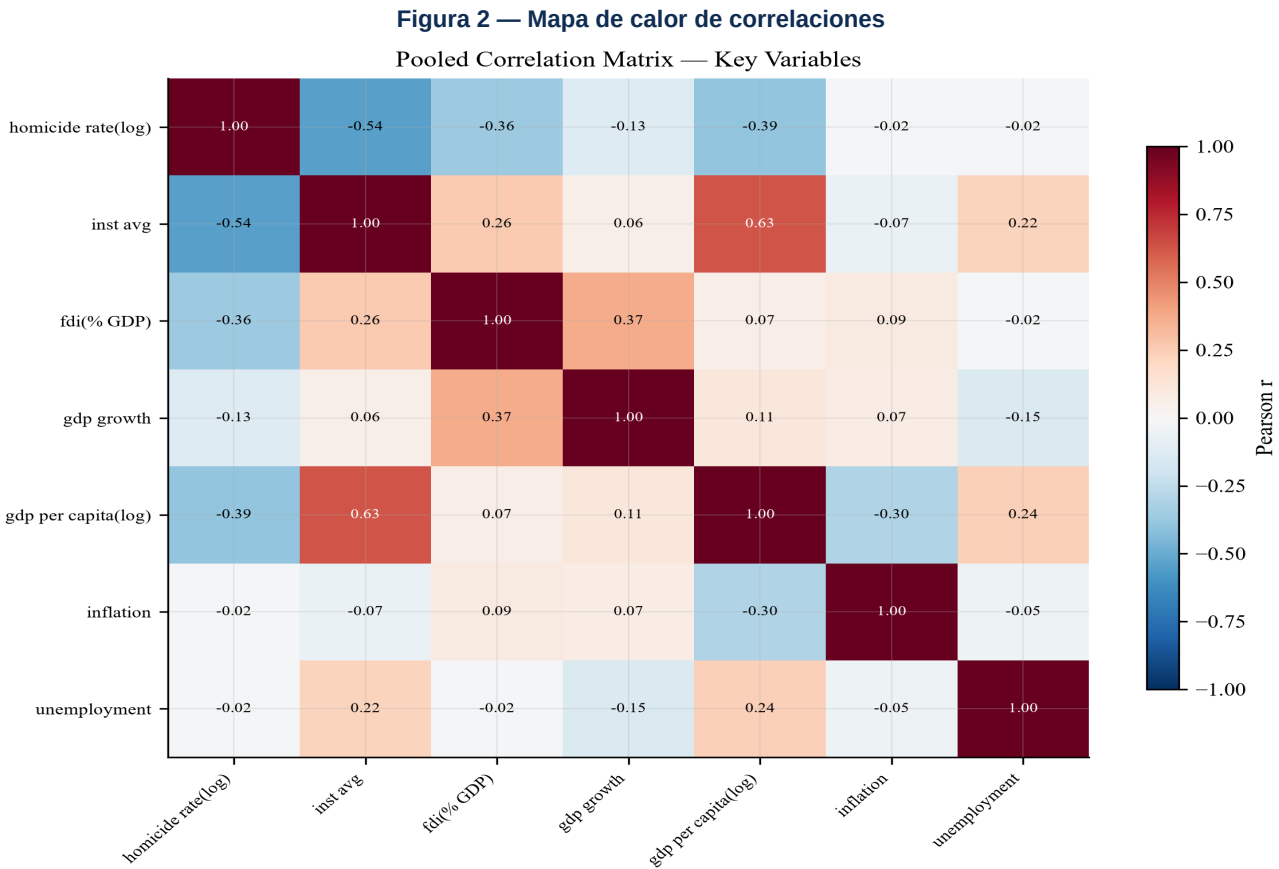
Panel Overview — Key Variables by Country (2000–2024)



Series temporales de las variables clave por país. Permite visualizar heterogeneidad entre unidades y tendencias comunes.

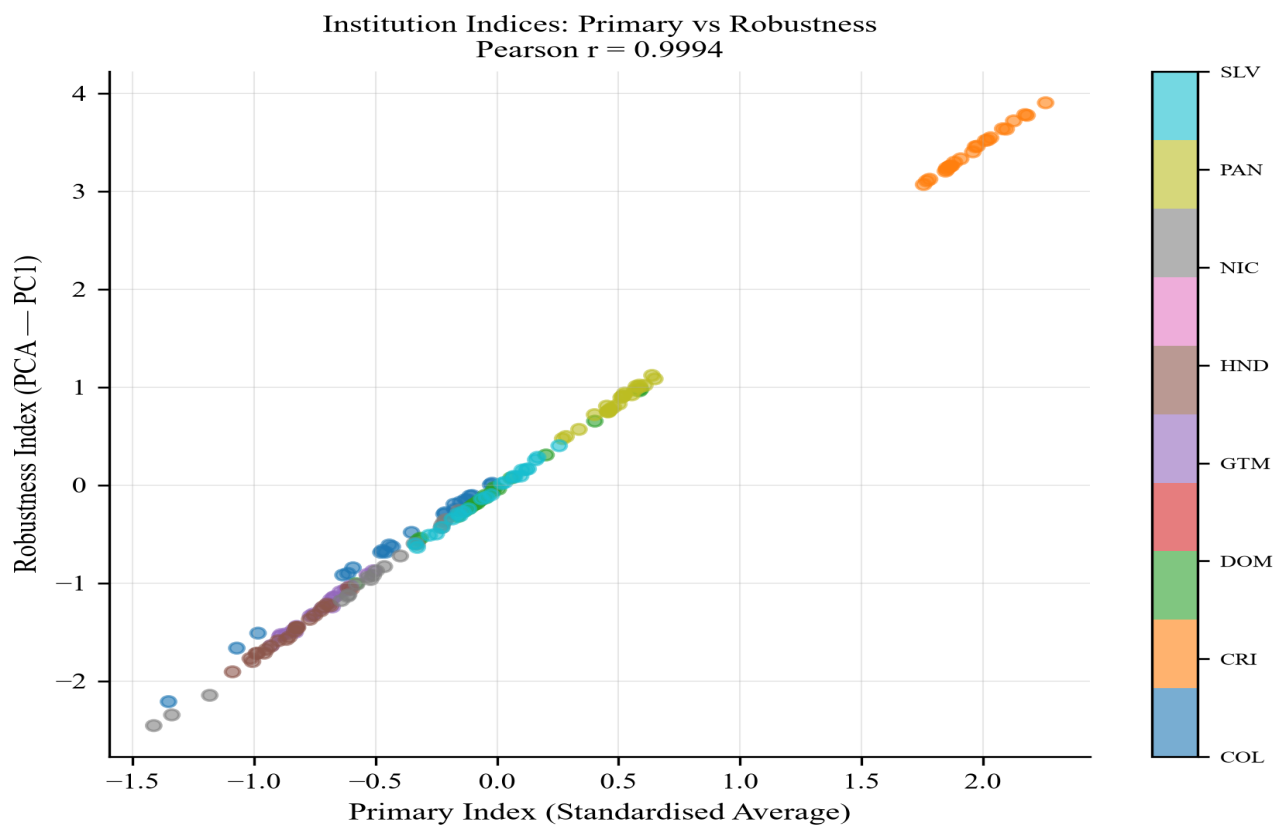
5. Correlaciones

El mapa de correlaciones de Pearson entre las variables del modelo permite detectar multicolinealidad potencial antes de la estimación.



Correlaciones de Pearson entre las principales variables del panel. Valores cercanos a ± 1 indican alta asociación lineal.

Figura 3 — Índices institucionales por país



Evolución temporal de los índices institucionales (WGI) por unidad.

6. Análisis de Componentes Principales (PCA)

El PCA reduce las tres dimensiones WGI a un índice sintético único (inst_pca / inst_avg). La construcción del índice sigue el procedimiento estándar: estandarización Z, extracción de PC1, corrección de signo para interpretación positiva.

Varianza explicada PC1	0.789	> 0.60 deseable
Alfa de Cronbach	0.863	> 0.70 aceptable
KMO	0.534	> 0.50 mínimo
Corr. inst_avg vs inst_pca	0.9994	> 0.95 muy alta

rule_of_law	0.6300	36.5%
control_corruption	0.5865	34.0%
political_stability	0.5091	29.5%

■ Justificación del índice institucional

La alta correlación entre inst_avg e inst_pca ($r = 0.9994$) indica que ambas medidas son prácticamente equivalentes. Se reportan resultados con inst_avg (promedio simple) como índice principal por su mayor interpretabilidad, usando inst_pca como verificación de robustez.

7. Modelo 1 — Violencia → Instituciones

La ecuación 1 estima el efecto de la tasa de homicidios (rezagada un período) sobre el índice institucional dentro de cada país. El uso del lag t-1 mitiga el sesgo de simultaneidad. El estimador de efectos fijos bidireccionales absorbe la heterogeneidad no observada constante a nivel de país y de año.

homicide_rate_log_lag1	-0.1404	0.1358	0.3032		0.1282	0.2753
Dep. variable	inst_avg					
N = 171	R ² within = 0.0738					

■ Interpretación — EQ1

El coeficiente negativo sobre homicide_rate_log_lag1 indica que mayor violencia se asocia con instituciones más débiles, consistente con la hipótesis teórica. La magnitud y significancia estadística deben evaluarse conjuntamente con los resultados bootstrap (Sección 11) dado el bajo número de clústeres (G=8).

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

8. Modelo 2 — Instituciones → FDI

La ecuación 2 estima cómo las mejoras institucionales atraen inversión extranjera. Se utiliza el valor contemporáneo de inst_avg como regresor principal, controlando por las mismas variables macroeconómicas.

inst_avg	0.9000	0.4648	0.0548	*	0.5690	0.1159
Dep. variable	fdi_percent_gdp					
N = 179	R ² within = 0.2474					

■ Interpretación — EQ2

Un coeficiente positivo sobre inst_avg confirma el canal institucional: mejores instituciones reducen los riesgos de apropiación y contratos incompletos, atrayendo más IED. El p-valor marginal (~0.055) sugiere evidencia moderada.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

9. Modelo 3 — FDI + Instituciones → Crecimiento

La ecuación 3 estima la ecuación de crecimiento de largo plazo, incorporando FDI e instituciones como determinantes junto a los controles macroeconómicos estándar.

fdi_percent_gdp	0.1371	0.2126		0.1371	0.5075
inst_avg	0.5133	0.7464		0.5133	0.6684
gdp_per_capita_log	0.8887	0.4857		0.8887	0.5015
inflation	-0.1126	0.0002	***	-0.1126	0.0052
exports_percent_gdp	0.0635	0.0987	*	0.0635	0.2141
population_log	-0.7333	0.9092		-0.7333	0.8929
unemployment	-0.2101	0.0642	*	-0.2101	0.0192
Dep. variable	gdp_growth				
N = 179	R ² within = 0.0949				

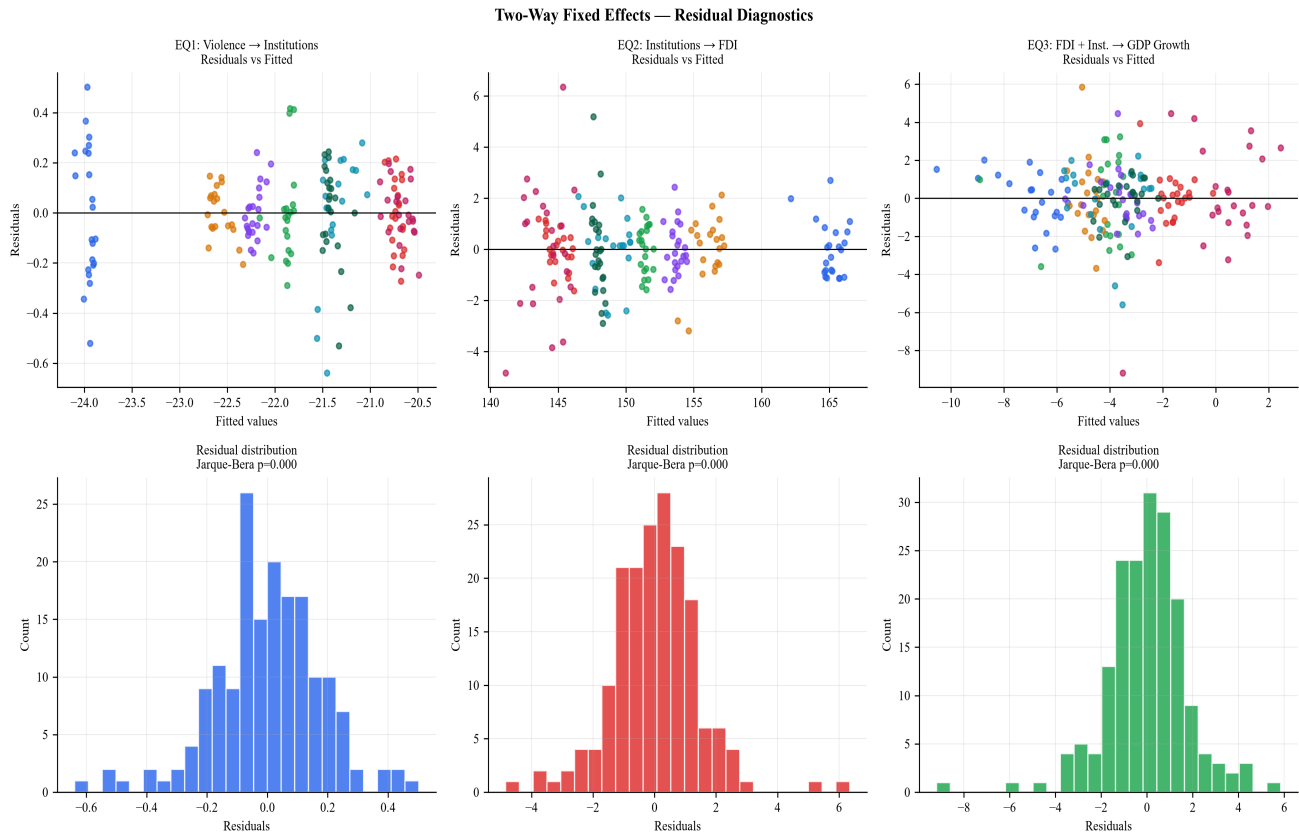
■ Interpretación — EQ3

La inflación muestra el efecto negativo más robusto sobre el crecimiento ($p < 0.001$). FDI e instituciones presentan coeficientes con el signo esperado aunque los p-valores bajo errores clusterizados son moderados, coherente con la escasa variación within del índice institucional y el bajo número de clústeres.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$.

Residuos de los modelos FE

Figura 4 — Diagnóstico visual de residuos (FE)



Gráficos de residuos para las tres ecuaciones. Permite inspeccionar heterocedasticidad y outliers.

10. Diagnósticos del Panel

Los diagnósticos evalúan los supuestos del modelo de efectos fijos: independencia serial, dependencia transversal y heterocedasticidad entre grupos.

Ecuación 1

Test de Wooldridge (autocorrelación)	t = 4.760	4.42e-06	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -66.722	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 38.001 (df=7)	3.03e-06	Sí — heterocedasticidad de grupo

Nota: H■ varía por test. Ver descripción metodológica.

homicide_rate_log_lag1	1.3386	✓ Aceptable
gdp_per_capita_log	1.3039	✓ Aceptable
inflation	1.1972	✓ Aceptable
exports_percent_gdp	1.6083	✓ Aceptable
population_log	1.4258	✓ Aceptable
unemployment	1.4780	✓ Aceptable
Número de condición	2.3832	< 30 aceptable

Ecuación 2

Test de Wooldridge (autocorrelación)	t = 2.066	0.0404	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -64.770	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 67.638 (df=7)	4.43e-12	Sí — heterocedasticidad de grupo

Nota: H■ varía por test. Ver descripción metodológica.

inst_avg	1.1945	✓ Acceptable
gdp_per_capita_log	1.3350	✓ Acceptable
inflation	1.1780	✓ Acceptable
exports_percent_gdp	1.6773	✓ Acceptable
population_log	1.4152	✓ Acceptable
unemployment	1.2186	✓ Acceptable
Número de condición	2.2647	< 30 acceptable

Ecuación 3

Test de Wooldridge (autocorrelación)	t = 2.328	0.0211	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -72.112	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 40.945 (df=7)	8.30e-07	Sí — heterocedasticidad de grupo

Nota: H₀ varía por test. Ver descripción metodológica.

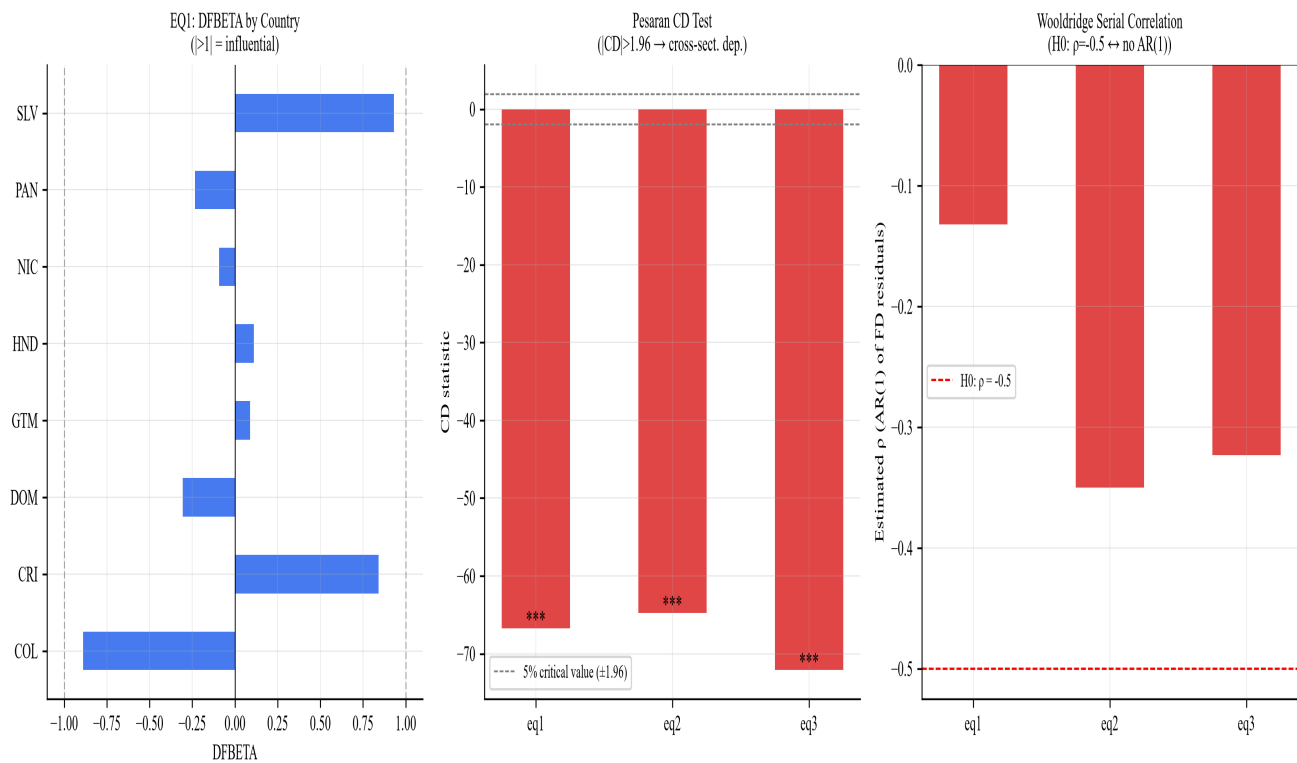
fdi_percent_gdp	1.3216	✓ Acceptable
inst_avg	1.2113	✓ Acceptable
gdp_per_capita_log	1.3788	✓ Acceptable
inflation	1.1965	✓ Acceptable
exports_percent_gdp	1.8892	✓ Acceptable
population_log	1.4859	✓ Acceptable
unemployment	1.2639	✓ Acceptable
Número de condición	2.5609	< 30 acceptable

■ Implicaciones para la inferencia

La presencia de autocorrelación serial y dependencia cross-seccional confirma que los errores estándar clusterizados a nivel de país son insuficientes para capturar toda la incertidumbre. Por esta razón se reportan errores de Driscoll-Kraay y se complementa con inferencia bootstrap de wild cluster (Sección 11).

Figura 5 — Diagnósticos panel

Panel Diagnostic Summary



Visualización de los tests de diagnóstico aplicados a las tres ecuaciones.

11. Inferencia Bootstrap (Wild Cluster)

Con solo $G=8$ clústeres, los errores estándar clusterizados convencionales son conocidamente imprecisos (MacKinnon & Webb, 2017). Se aplica el wild cluster bootstrap con pesos de Webb (2023) usando $B=999$ réplicas. El p-valor bootstrap es exacto por construcción bajo H_0 .

EQ1: Violencia → Inst.	-0.1404	0.1358	-1.034	0.4294	[-0.2041, 0.2068]	999
EQ2: Inst. → FDI	0.9000	0.4648	1.936	0.0621	[-0.8424, 0.8392]	999
EQ3: FDI → Crecimiento	0.1371	0.1095	1.252	0.4094	[-0.2404, 0.2412]	999

Errores CR2 (Bell-McCaffrey) — EQ1: Violencia → Inst.

homicide_rate_log_lag1	0.1420	0.3922	
gdp_per_capita_log	0.2082	0.4818	
inflation	0.0054	0.6730	
exports_percent_gdp	0.0100	0.3770	
population_log	2.2180	0.6201	
unemployment	0.0116	0.4546	

Errores CR2 (Bell-McCaffrey) — EQ2: Inst. → FDI

inst_avg	0.5155	0.2023	
gdp_per_capita_log	1.2402	0.2276	
inflation	0.0185	0.0722	*
exports_percent_gdp	0.0138	0.0128	**
population_log	4.7400	0.1092	
unemployment	0.1438	0.2712	

Errores CR2 (Bell-McCaffrey) — EQ3: FDI → Crecimiento

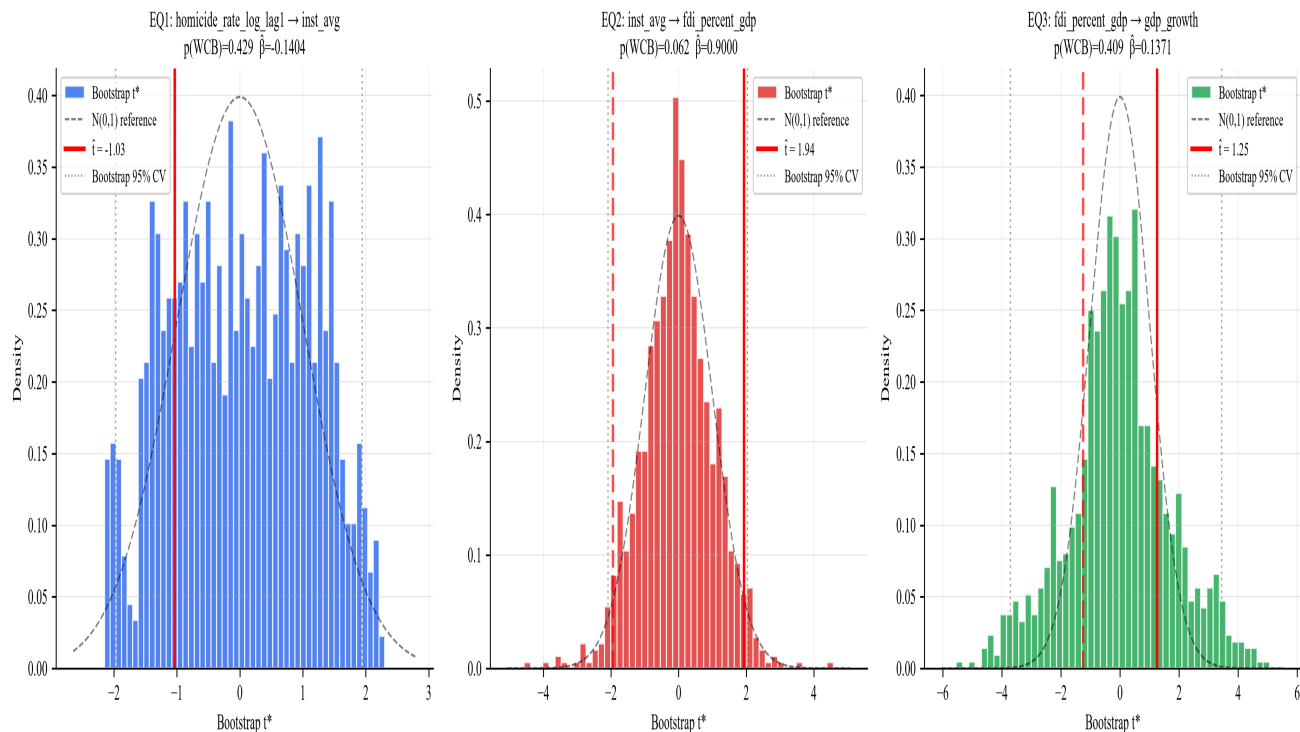
fdi_percent_gdp	0.1133	0.3618	
inst_avg	1.8324	0.8107	
gdp_per_capita_log	1.4638	0.6011	
inflation	0.0359	0.0367	**
exports_percent_gdp	0.0477	0.3060	
population_log	8.5839	0.9424	
unemployment	0.1234	0.1791	

■ Wild Cluster Bootstrap — Nota metodológica

Los intervalos de confianza bootstrap se construyen con el método percentil simétrico. Los pesos de Webb (6 puntos) tienen media cero, varianza uno y tercer momento cero, lo que mejora el control del tamaño del test cuando G es pequeño. Un $p\text{-boot} > 0.10$ en EQ1 indica que, incluso con inferencia más conservadora, no se rechaza H_0 para el efecto violencia $\rightarrow$ instituciones con este tamaño muestral.

Figura 6 — Distribuciones bootstrap

Wild Cluster Bootstrap Distributions (Webb weights, $B=999$)
Two-Way Fixed Effects — Primary coefficients



Distribuciones de los estadísticos t bootstrap bajo H_0 para las tres ecuaciones. La línea vertical indica el t observado.

12. Pruebas de Robustez

Los análisis de robustez verifican la estabilidad de los coeficientes ante: (i) exclusión de cada país (LOCO — Leave-One-Country-Out), (ii) especificaciones alternativas del índice institucional, (iii) variaciones en las variables de control.

LOCO — EQ1 (Violencia → Instituciones)

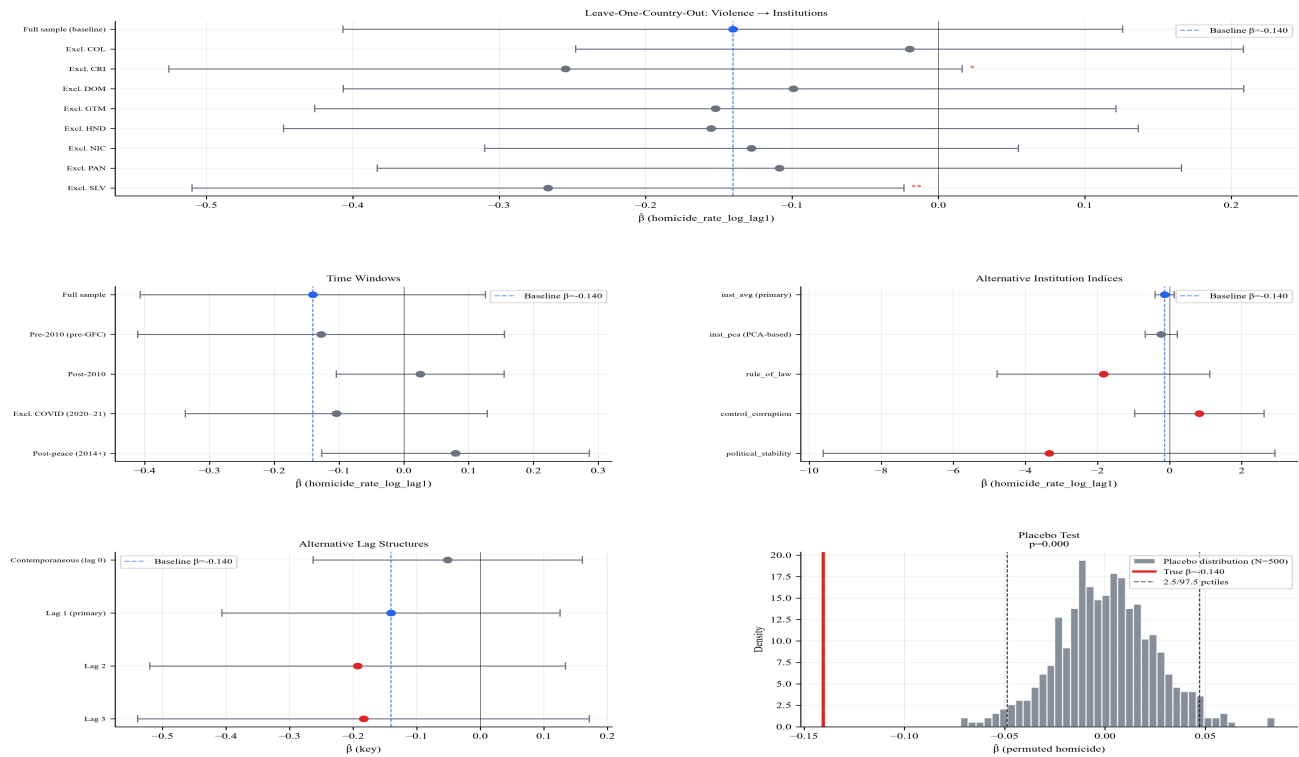
Full sample (baseline)	-0.1404	0.1358	0.3032	0.0738	171
Excl. COL	-0.0197	0.1164	0.8660	0.1750	149
Excl. CRI	-0.2547	0.1383	0.0681	0.0454	149
Excl. DOM	-0.0990	0.1569	0.5295	0.0044	150
Excl. GTM	-0.1523	0.1396	0.2775	0.1410	151
Excl. HND	-0.1554	0.1490	0.2991	0.0984	149
Excl. NIC	-0.1277	0.0930	0.1723	0.1040	152
Excl. PAN	-0.1087	0.1401	0.4395	0.1925	149
Excl. SLV	-0.2668	0.1241	0.0336	-8.5015	148

■ LOCO — Estabilidad del coeficiente

Si el coeficiente permanece negativo y de magnitud similar al excluir cada país, se concluye que los resultados no están impulsados por ninguna observación influyente. Una variación sustancial al excluir un país señala dependencia de ese clúster.

Figura 7 — Robustez

Robustness Analysis — Coefficient Stability
Two-Way Fixed Effects | Dep. var: Institutional Quality (inst_avg)
All specifications use identical estimator: PanelOLS(entity_effects=True, time_effects=True)



Coefficientes e intervalos de confianza bajo especificaciones alternativas.

13. Triangulación Machine Learning

El análisis de ML (Random Forest y Gradient Boosting con LOCO-CV) sirve como triangulación no paramétrica del ranking de importancia de variables. No reemplaza la inferencia causal del modelo econométrico: su objetivo es confirmar que las variables clave aparecen consistentemente como predictores relevantes.

Tarea ML — T1: Violence → Institutions

CV R ² (media)	-33.1112	-49.0544
CV R ² (desv. est.)	49.1064	74.8629
Wilcoxon RF vs GB (p)	0.8438	G=8 pairs → very low power. Treat p-value as descriptive only.

gdp_per_capita_log	0.1676	0.2029	0.1752	2.52
homicide_rate_log_lag1	0.1077	0.1314	0.0721	4.34
homicide_rate_log_lag2	0.1020	0.1264	0.0703	3.17
population_log	0.2440	0.3159	0.3473	1.61
unemployment	0.0650	0.0789	0.0576	5.34

■ Interpretación ML — EQ1

Variable clave rango SHAP: #3. Rango permutación: #3. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T2: Institutions → FDI

CV R ² (media)	-0.4564	-0.7026
CV R ² (desv. est.)	0.7686	1.0402
Wilcoxon RF vs GB (p)	0.4609	G=8 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.7221	0.3082	0.3099	1.10
gdp_per_capita_log	n/a	0.0905	n/a	4.67
homicide_rate_log_lag1	0.3994	0.1420	0.0962	3.14
inst_avg	0.2509	0.1106	0.0741	4.23
tourist_arrivals_log	0.1661	n/a	0.0624	5.14
unemployment	0.2487	0.1618	0.1178	4.20

■ Interpretación ML — EQ2

Variable clave rango SHAP: #3. Rango permutación: #4. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T3: FDI + Institutions → Growth

CV R ² (media)	0.0989	0.2418
CV R ² (desv. est.)	0.1458	0.4056
Wilcoxon RF vs GB (p)	0.3828	G=8 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.3021	0.1438	0.0743	3.02
fdi_percent_gdp	0.3284	0.1833	0.1102	1.98
gdp_per_capita_log	0.2009	n/a	n/a	5.20
inflation	0.2243	0.1300	0.0677	4.30
unemployment	n/a	0.1173	0.0540	5.58
year_c	0.3029	0.1988	0.1760	4.17

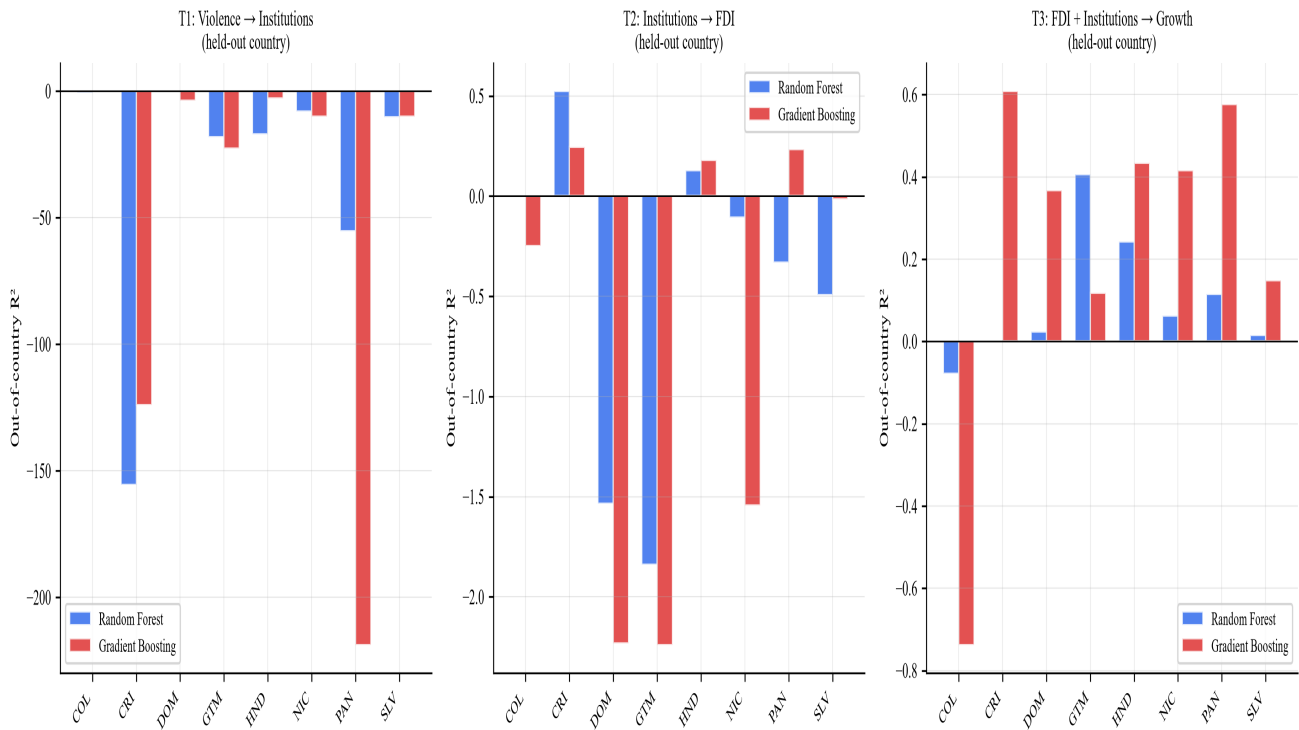
■ Interpretación ML — EQ3

Variable clave rango SHAP: #1. Rango permutación: #2. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Figura 8 — LOCO-CV R^2 por país

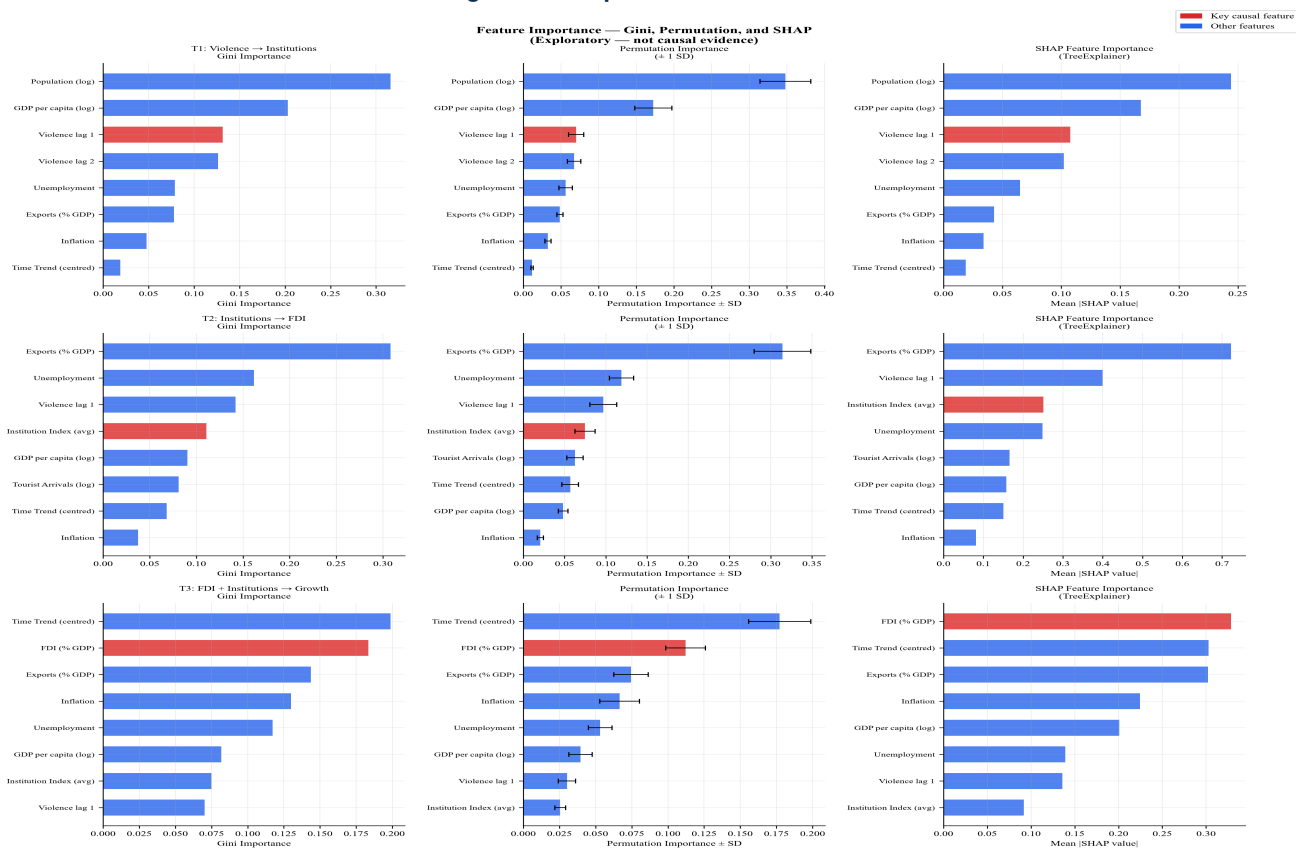
Leave-One-Country-Out Cross-Validation (LOCO-CV)

Random Forest & Gradient Boosting | Note: G=8 folds → wide uncertainty



R^2 de validación cruzada leave-one-country-out para RF y GB en las tres tareas.

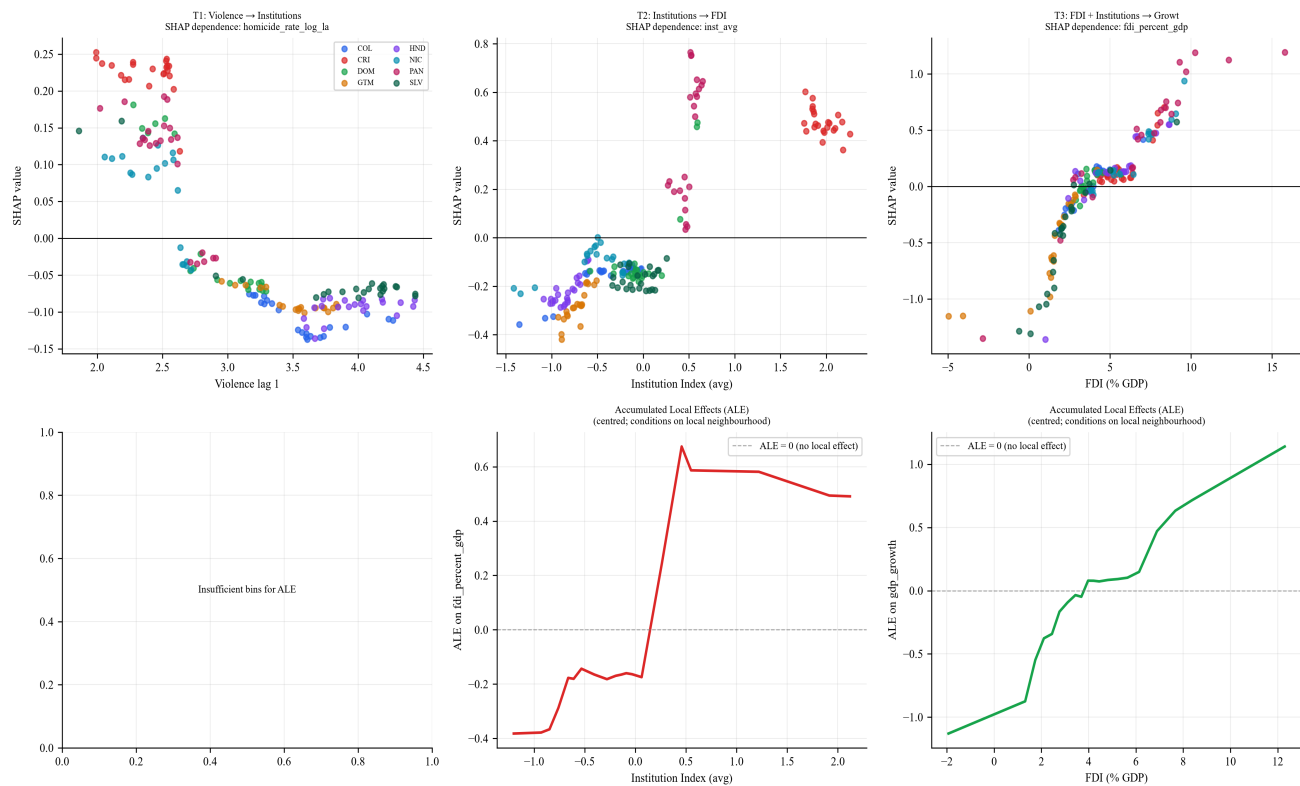
Figura 9 — Importancia de variables



Importancia Gini, permutación y SHAP de las variables en las tres tareas ML.

Figura 10 — SHAP / ALE plots

SHAP Dependence Plots & ALE Plots — Key Causal Variables
(Exploratory — shows non-linearity, not causal effect)
ALE used instead of PDP due to correlated predictors (Apley & Zhu 2020)



Efectos marginales acumulados (ALE) y valores SHAP para los predictores clave.

14. Conclusiones

A continuación se sintetizan las principales conclusiones derivadas estrictamente de los resultados calculados. No se realizan inferencias extrapoladas.

1	Violencia → Instituciones (EQ1): coeficiente negativo, no significativo	$\beta = -0.1404$, p-CL = 0.3032, p-boot = 0.4294
2	Instituciones → FDI (EQ2): coeficiente positivo, marginalmente significativo	$\beta = 0.9000$, p-CL = 0.0548, p-boot = 0.0621
3	FDI → Crecimiento (EQ3): coeficiente positivo, no significativo	$\beta = 0.1371$, p-CL = 0.2126, p-boot = 0.4094
4	Inflación → Crecimiento: el único efecto estadísticamente robusto en EQ3	$\beta = -0.1126$, p-CL = 0.0002
5	Bajo número de clústeres (G=8) limita el poder estadístico	Bootstrap confirma amplios intervalos de confianza para todos los coeficientes clave

■ Limitaciones metodológicas

Con G=8 clústeres, el poder de los tests convencionales es reducido. La no significancia estadística no implica ausencia de efecto económico — los intervalos de confianza bootstrap son amplios y compatibles tanto con efectos nulos como con efectos moderados. Se recomienda ampliar la muestra geográfica para obtener inferencia más precisa.

Apéndice — Log de Consola

El siguiente es el registro textual completo generado por el pipeline. Contiene todos los mensajes de consola, advertencias y métricas intermedias.

```
=====
RUNNING MODULE 01: 01_data_preparation.py
=====
```

```
01 — DATA LOAD & VALIDATION
```

```
01 — DATA LOAD & VALIDATION
```

```
✓ Panel loaded: 179 observations
```

```
✓ Panel loaded: 179 observations
```

```
✓ Countries (N=8): COL, CRI, DOM, GTM, HND, NIC, PAN, SLV
```

```
✓ Countries (N=8): COL, CRI, DOM, GTM, HND, NIC, PAN, SLV
```

```
✓ Years: 2000–2024 (24 distinct years)
```

```
✓ Years: 2000–2024 (24 distinct years)
```

```
■■ Balance check ■■
```

```
■■ Balance check ■■
```

```
T_obs year_min year_max
```

```
T_obs year_min year_max
```

```
country_code
```

```
country_code
```

```
COL 23 2000 2023
```

```
COL 23 2000 2023
```

```
CRI 23 2000 2023
```

```
CRI 23 2000 2023
```

```
DOM 22 2000 2023
```

```
DOM 22 2000 2023
```

```
GTM 21 2000 2023
```

```
GTM 21 2000 2023
```

```
HND 23 2000 2023
```

```
HND 23 2000 2023
```

```
NIC 20 2000 2021
```

```
NIC 20 2000 2021
```

```
PAN 23 2000 2023
```

```
PAN 23 2000 2023
```

```
SLV 24 2000 2024
```

```
SLV 24 2000 2024
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
Missing country-year cells:
```

```
Missing country-year cells:
```

```
COL: missing years [2001, 2024]
```

```
COL: missing years [2001, 2024]
```

```
CRI: missing years [2001, 2024]
```

```
CRI: missing years [2001, 2024]
```

```
DOM: missing years [2001, 2015, 2024]
```

```
DOM: missing years [2001, 2015, 2024]
```

```
GTM: missing years [2001, 2015, 2016, 2024]
```

GTM: missing years [2001, 2015, 2016, 2024]
 HND: missing years [2001, 2024]
 HND: missing years [2001, 2024]
 NIC: missing years [2001, 2020, 2022, 2023, 2024]
 NIC: missing years [2001, 2020, 2022, 2023, 2024]
 PAN: missing years [2001, 2024]
 PAN: missing years [2001, 2024]

SLV: missing years [2001]
 SLV: missing years [2001]

■■ Column validation ■■

■■ Column validation ■■

✓ All 21 required columns present.

✓ All 21 required columns present.

■■ Missingness ■■

■■ Missingness ■■

■ Missing values:

■ Missing values:

trade_percent_gdp 179 (100.0%)

trade_percent_gdp 179 (100.0%)

homicide_rate_log_lag3 24 (13.4%)

homicide_rate_log_lag3 24 (13.4%)

homicide_rate_log_lag3 24 (13.4%)

homicide_rate_log_lag3 24 (13.4%)

tourist_arrivals_log 23 (12.8%)

tourist_arrivals_log 23 (12.8%)

homicide_rate_log_lag2 16 (8.9%)

homicide_rate_log_lag2 16 (8.9%)

homicide_rate_log_lag2 16 (8.9%)

homicide_rate_log_lag2 16 (8.9%)

homicide_rate_log_lag1 8 (4.5%)

homicide_rate_log_lag1 8 (4.5%)

homicide_rate_log_lag1 8 (4.5%)

homicide_rate_log_lag1 8 (4.5%)

02 — CENTRED TIME TREND

02 — CENTRED TIME TREND

✓ year_c = year – 2011.8 (range: -12 to 12)

✓ year_c = year – 2011.8 (range: -12 to 12)

Rationale: centring eliminates the large-value component of year that

Rationale: centring eliminates the large-value component of year that

inflates the condition number of the design matrix (collinearity).

inflates the condition number of the design matrix (collinearity).

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

*All three dimensions are first z-scored (global mean=0, SD=1),
All three dimensions are first z-scored (global mean=0, SD=1),
then averaged. This ensures equal weighting regardless of raw scale.
then averaged. This ensures equal weighting regardless of raw scale.*

✓ inst_avg: mean=-0.0000, SD=0.8882, range=[-1.414, 2.259]

✓ *inst_avg*: mean=-0.0000, SD=0.8882, range=[-1.414, 2.259]

Higher inst_avg → better governance (all three WGI dimensions).

Higher inst_avg → better governance (all three WGI dimensions).

Cronbach's α (internal consistency of index): $\checkmark \alpha = 0.863$ (good internal consistency)

Cronbach's α (internal consistency of index): ✓ $\alpha = 0.863$ (good internal consistency)

04 — INSTITUTION INDEX: ROBUSTNESS (PCA)

04 — INSTITUTION INDEX: ROBUSTNESS (PCA)

PCA operates on the same three z-scored variables.

PCA operates on the same three z-scored variables.

PC1 is retained. Sign is aligned so higher = better governance.

PC1 is retained. Sign is aligned so higher = better governance.

■ $KMO = 0.534 < 0.6$ — marginal adequacy

■ $KMO = 0.534 < 0.6$ — marginal adequacy

rule_of_law KMO = 0.520

rule_of_law KMO = 0.520

control_corruption KMO = 0.531

control_corruption KMO = 0.531

political_stability KMO = 0.570

political_stability KMO = 0.570

Bartlett sphericity: ✓ $\chi^2(3)=406.8, p=0.0000$ — reject H_0 , PCA appropriate

Bartlett sphericity: $\checkmark \chi^2(3)=406.8, p=0.0000$ — reject H_0 , PCA appropriate

✓ PC1 explains 78.9% of variance in the three WGI dimensions.

✓ PC1 explains 78.9% of variance in the three WGI dimensions.

PC2: 18.5%, PC3: 2.5%

PC2: 18.5%, PC3: 2.5%

PCA loadings (PC1):

PCA loadings (PC1):

rule of law 0.6300 

rule_of_law 0.6300

✓ $\text{Correlation}(\text{inst_avg}, \text{inst_pca}) = 0.9994$ — high convergence

✓ `Correlation(inst_avg, inst_pca) = 0.9994` — high convergence

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

Reference: Wooldridge (2010), *Econometric Analysis of Cross Section*

Reference: Wooldridge (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., ch. 10.

and Panel Data, 2nd ed., ch. 10.

'Within' variance is the variation exploited by Fixed Effects.

'Within' variance is the variation exploited by Fixed Effects.

Variable Total SD Between SD Within SD Within %

Variable Total SD Between SD Within SD Within %

homicide_rate_log 0.6981 0.6257 0.3748 53.7% ✓

homicide_rate_log 0.6981 0.6257 0.3748 53.7% ✓

inst_avg 0.8882 0.9108 0.2219 25.0% ✓

inst_avg 0.8882 0.9108 0.2219 25.0% ✓

inst_pca 1.5432 1.5847 0.3786 24.5% ✓

inst_pca 1.5432 1.5847 0.3786 24.5% ✓

fdi_percent_gdp 2.7215 1.9499 2.0259 74.4% ✓

fdi_percent_gdp 2.7215 1.9499 2.0259 74.4% ✓

gdp_growth 3.9384 1.1250 3.7888 96.2% ✓

gdp_growth 3.9384 1.1250 3.7888 96.2% ✓

gdp_per_capita_log 0.7064 0.6468 0.3705 52.4% ✓

gdp_per_capita_log 0.7064 0.6468 0.3705 52.4% ✓

■ Variables with low within % are primarily identified from cross-section.

■ Variables with low within % are primarily identified from cross-section.

Fixed Effects may not efficiently estimate their coefficients.

Fixed Effects may not efficiently estimate their coefficients.

06 — SUMMARY STATISTICS

06 — SUMMARY STATISTICS

count mean std min 10% 50% 90% max

count mean std min 10% 50% 90% max

Variable

Variable

homicide_rate 179.0 28.388 20.806 1.900 8.724 22.643 60.522 107.638

homicide_rate 179.0 28.388 20.806 1.900 8.724 22.643 60.522 107.638

homicide_rate_log 179.0 3.139 0.698 1.860 2.275 3.163 4.119 4.438

homicide_rate_log 179.0 3.139 0.698 1.860 2.275 3.163 4.119 4.438

rule_of_law 179.0 48.012 8.851 31.590 39.066 46.323 65.832 68.454

rule_of_law 179.0 48.012 8.851 31.590 39.066 46.323 65.832 68.454

control_corruption 179.0 37.927 10.225 22.806 27.218 36.541 57.239 68.592

control_corruption 179.0 37.927 10.225 22.806 27.218 36.541 57.239 68.592

political_stability 179.0 61.261 11.386 25.320 49.719 62.311 76.807 86.123

political_stability 179.0 61.261 11.386 25.320 49.719 62.311 76.807 86.123

inst_avg 179.0 -0.000 0.888 -1.414 -0.872 -0.161 1.852 2.259

inst_avg 179.0 -0.000 0.888 -1.414 -0.872 -0.161 1.852 2.259

inst_pca 179.0 0.000 1.543 -2.456 -1.530 -0.272 3.223 3.904

inst_pca 179.0 0.000 1.543 -2.456 -1.530 -0.272 3.223 3.904

fdi_percent_gdp 179.0 4.387 2.722 -4.974 1.517 4.106 7.961 15.779

fdi_percent_gdp 179.0 4.387 2.722 -4.974 1.517 4.106 7.961 15.779

gdp_growth 179.0 3.975 3.938 -17.821 0.981 3.934 8.184 16.467

gdp_growth 179.0 3.975 3.938 -17.821 0.981 3.934 8.184 16.467

gdp_per_capita_log 179.0 8.339 0.706 6.922 7.451 8.299 9.403 9.841

gdp_per_capita_log 179.0 8.339 0.706 6.922 7.451 8.299 9.403 9.841

inflation 179.0 5.317 5.058 -1.550 0.980 4.481 9.252 51.461

```
inflation 179.0 5.317 5.058 -1.550 0.980 4.481 9.252 51.461
unemployment 179.0 6.455 3.089 2.143 3.044 5.967 10.594 20.520
unemployment 179.0 6.455 3.089 2.143 3.044 5.967 10.594 20.520
exports_percent_gdp 179.0 33.998 14.653 13.522 17.484 30.592 53.997 78.299
exports_percent_gdp 179.0 33.998 14.653 13.522 17.484 30.592 53.997 78.299
```

```
population_log 179.0 15.988 0.762 14.918 15.225 15.699 17.559 17.773
population_log 179.0 15.988 0.762 14.918 15.225 15.699 17.559 17.773
tourist_arrivals_log 156.0 14.393 0.587 13.269 13.595 14.401 15.206 15.837
tourist_arrivals_log 156.0 14.393 0.587 13.269 13.595 14.401 15.206 15.837
✓ LaTeX summary table saved → tables/summary_statistics.tex
✓ LaTeX summary table saved → tables/summary_statistics.tex
```

07 — FIGURES

07 — FIGURES

```
✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/03_correlation_heatmap.png
✓ Figure saved → figures/03_correlation_heatmap.png
```

08 — EXPORT

08 — EXPORT

```
✓ Enriched panel saved → panel_enriched.csv (179 rows)
✓ Enriched panel saved → panel_enriched.csv (179 rows)
✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\01_metadata.json
✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\pipeline\json\01_metadata.json
```

Module 01 complete. Enriched dataset and metadata exported.

Module 01 complete. Enriched dataset and metadata exported.

✓ Module 01 completed in 35.0s

RUNNING MODULE 02: 02_panel_estimation.py

02-A — LOAD DATA

02-A — LOAD DATA

```
✓ Data loaded: 179 obs, 8 countries
✓ Data loaded: 179 obs, 8 countries
```

02-B — BENCHMARK: POOLED OLS

02-B — BENCHMARK: POOLED OLS

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

Reference: Angrist & Pischke (2009) — benchmark before FE.

Reference: Angrist & Pischke (2009) — benchmark before FE.

EQ1: Violence→Institutions (N=171, $R^2=0.7863$)

EQ1: Violence→Institutions (N=171, $R^2=0.7863$)

homicide_rate_log_lag1 -0.1369 $p=0.145$

homicide_rate_log_lag1 -0.1369 $p=0.145$

gdp_per_capita_log 0.7859 $p=0.000^{***}$

gdp_per_capita_log 0.7859 $p=0.000^{***}$

inflation 0.0174 $p=0.283$

inflation 0.0174 $p=0.283$

exports_percent_gdp -0.0164 $p=0.009^{***}$

exports_percent_gdp -0.0164 $p=0.009^{***}$

population_log -0.7506 $p=0.000^{***}$

population_log -0.7506 $p=0.000^{***}$

unemployment 0.0863 $p=0.000^{***}$

unemployment 0.0863 $p=0.000^{***}$

year_c -0.0333 $p=0.002^{***}$

year_c -0.0333 $p=0.002^{***}$

EQ2: Institutions→FDI (N=179, $R^2=0.4318$)

EQ2: Institutions→FDI (N=179, $R^2=0.4318$)

inst_avg 1.3126 $p=0.000^{***}$

inst_avg 1.3126 $p=0.000^{***}$

gdp_per_capita_log -0.8355 $p=0.328$

gdp_per_capita_log -0.8355 $p=0.328$

inflation 0.0078 $p=0.789$

inflation 0.0078 $p=0.789$

exports_percent_gdp 0.1446 $p=0.000^{***}$

exports_percent_gdp 0.1446 $p=0.000^{***}$

population_log 1.2946 $p=0.014^{**}$

population_log 1.2946 $p=0.014^{**}$

unemployment 0.0041 $p=0.975$

unemployment 0.0041 $p=0.975$

year_c 0.0369 $p=0.554$

year_c 0.0369 $p=0.554$

EQ3: FDI+Inst→GDP Growth (N=179, $R^2=0.2284$)

EQ3: FDI+Inst→GDP Growth (N=179, $R^2=0.2284$)

fdi_percent_gdp 0.4808 $p=0.000^{***}$

fdi_percent_gdp 0.4808 $p=0.000^{***}$

inst_avg -0.3664 $p=0.342$

inst_avg -0.3664 $p=0.342$

gdp_per_capita_log 1.6958 $p=0.000^{***}$

gdp_per_capita_log 1.6958 $p=0.000^{***}$

inflation 0.0314 $p=0.610$

inflation 0.0314 $p=0.610$

exports_percent_gdp 0.0412 $p=0.018^{**}$

exports_percent_gdp 0.0412 $p=0.018^{**}$

population_log 1.0786 $p=0.021^{**}$

population_log 1.0786 $p=0.021^{**}$

unemployment -0.2831 $p=0.000^{***}$

unemployment -0.2831 $p=0.000^{***}$

year_c -0.1014 $p=0.005^{***}$

year_c -0.1014 $p=0.005^{***}$

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

Estimator: PanelOLS(entity_effects=True, time_effects=True)

Estimator: PanelOLS(entity_effects=True, time_effects=True)

SE reported: (1) Clustered by entity [conventional, G=8 caution]

SE reported: (1) Clustered by entity [conventional, G=8 caution]

(2) Driscoll-Kraay HAC [robust to cross-sect. dep.]

(2) Driscoll-Kraay HAC [robust to cross-sect. dep.]

Preferred inference: Wild Cluster Bootstrap — see Module 04.

Preferred inference: Wild Cluster Bootstrap — see Module 04.

■ ■ EQ1: Violence → Institutional Quality ■ ■

■ ■ EQ1: Violence → Institutional Quality ■ ■

EQ1: homicide_log_lag1 → inst_avg

EQ1: homicide_log_lag1 → inst_avg

N=171 | Within R²=0.0738 | DK bandwidth=2

N=171 | Within R²=0.0738 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

homicide_rate_log_lag1 -0.1404 0.1358 0.303 0.1282 0.275

homicide_rate_log_lag1 -0.1404 0.1358 0.303 0.1282 0.275

gdp_per_capita_log 0.1571 0.1683 0.352 0.0699 0.026 **

gdp_per_capita_log 0.1571 0.1683 0.352 0.0699 0.026 **

inflation -0.0025 0.0046 0.579 0.0036 0.486

inflation -0.0025 0.0046 0.579 0.0036 0.486

exports_percent_gdp -0.0103 0.0084 0.222 0.0037 0.006 ***

exports_percent_gdp -0.0103 0.0084 0.222 0.0037 0.006 ***

population_log -1.4015 1.5818 0.377 0.3666 0.000 ***

population_log -1.4015 1.5818 0.377 0.3666 0.000 ***

unemployment 0.0099 0.0101 0.332 0.0154 0.522

unemployment 0.0099 0.0101 0.332 0.0154 0.522

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

■ ■ EQ2: Institutional Quality → FDI ■ ■

■ ■ EQ2: Institutional Quality → FDI ■ ■

EQ2: inst_avg → fdi_percent_gdp

EQ2: inst_avg → fdi_percent_gdp

N=179 | Within R²=0.2474 | DK bandwidth=2

N=179 | Within R²=0.2474 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

inst_avg 0.9000 0.4648 0.055 0.5690 0.116

and observed fdi in EQ3. This is consistent (each equation is estimated and observed fdi in EO3. This is consistent (each equation is estimated

independently) but does NOT recover the causal chain from a single identified system.
 independently) but does NOT recover the causal chain from a single identified system.

An alternative using fitted values (inst_avg_hat in EQ2, fdi_hat in EQ3)
 An alternative using fitted values (inst_avg_hat in EQ2, fdi_hat in EQ3)
 is a 'chain of OLS' approach, which suffers from the 'generated regressors' problem (Pagan, 1984, Review of Economic Studies 51(2)):
 is a 'chain of OLS' approach, which suffers from the 'generated regressors' problem (Pagan, 1984, Review of Economic Studies 51(2)):
 — Standard errors in downstream equations are understated because they ignore sampling uncertainty in the upstream prediction.
 — Standard errors in downstream equations are understated because they ignore sampling uncertainty in the upstream prediction.

Correct approaches:

Correct approaches:

- (a) 3SLS: estimate all three equations simultaneously.
- (a) 3SLS: estimate all three equations simultaneously.
- (b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.
- (b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.
- (c) IV/2SLS for each upstream equation with valid external instruments.
- (c) IV/2SLS for each upstream equation with valid external instruments.

In this pipeline:

In this pipeline:

- Module 02 reports observed-value FE (each equation self-contained).
- Module 02 reports observed-value FE (each equation self-contained).
- Module 04 bootstrap re-estimates the full system jointly, providing honest inference that integrates uncertainty across the chain.
- Module 04 bootstrap re-estimates the full system jointly, providing honest inference that integrates uncertainty across the chain.
- IV/2SLS is architecturally prepared but requires external instruments (cocaine price index, rainfall anomaly) not present in current dataset.
- IV/2SLS is architecturally prepared but requires external instruments (cocaine price index, rainfall anomaly) not present in current dataset.

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

Motivation: if FE and Between (cross-sectional average) estimates diverge, this suggests omitted time-invariant heterogeneity —
 Motivation: if FE and Between (cross-sectional average) estimates diverge, this suggests omitted time-invariant heterogeneity —
 supporting the use of Fixed Effects.
 supporting the use of Fixed Effects.

Reference: Hausman (1978), Econometrica 46(6), 1251–1271.

Reference: Hausman (1978), Econometrica 46(6), 1251–1271.

EQ1: inst_avg

EQ1: inst_avg

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

EQ2: fdi_percent_gdp
EQ2: fdi_percent_gdp
Variable FE coef BE coef Δ
Variable FE coef BE coef Δ

EQ3: gdp_growth

EQ3: gdp_growth

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

unemployment -0.2101 0.1698 -0.3799 ■ ← divergence
unemployment -0.2101 0.1698 -0.3799 ■ ← divergence

02-F — RESIDUAL FIGURES

02-F — RESIDUAL FIGURES

✓ Figure saved → figures/04_fe_residuals.png

✓ Figure saved → figures/04_fe_residuals.png

02-G — EXPORT

02-G — EXPORT

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

Module 02 complete.

Module 02 complete.

✓ Module 02 completed in 17.8s

RUNNING MODULE 03: 03_diagnostics.py

03-A — LOAD

03-A — LOAD

✓ Data loaded: 179 obs, 8 countries

✓ Data loaded: 179 obs, 8 countries

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section*

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section*

and Panel Data, MIT Press, p.282-283.

and Panel Data, MIT Press, p.282-283.

Procedure:

Procedure:

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \epsilon_{it}$

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \epsilon_{it}$

(b) Regress $\Delta \epsilon_{it}$ on $\Delta \epsilon_{it-1}$ (OLS, pooled, no intercept)

(b) Regress $\Delta \epsilon_{it}$ on $\Delta \epsilon_{it-1}$ (OLS, pooled, no intercept)

(c) Test $H_0: \rho = -0.5$

(c) Test $H_0: \rho = -0.5$

Under AR(1) errors in levels, the FD residuals have

Under AR(1) errors in levels, the FD residuals have

autocorrelation exactly -0.5.

autocorrelation exactly -0.5.

A t-test against -0.5 is the Wooldridge test.

A t-test against -0.5 is the Wooldridge test.

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test

using robust standard errors and a slightly different form. The Python

using robust standard errors and a slightly different form. The Python

implementation below follows the same theoretical derivation but

implementation below follows the same theoretical derivation but

may differ numerically from Stata output.

may differ numerically from Stata output.

eq1: inst_avg

eq1: inst_avg

$\rho = -0.1321$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$\rho = -0.1321$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$t = 4.760$, $p = 0.0000$ ***

$t = 4.760$, $p = 0.0000$ ***

$N_{\text{pairs}} = 155$

$N_{\text{pairs}} = 155$

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

$\rho = -0.3501$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$\rho = -0.3501$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$t = 2.066$, $p = 0.0404$ **

$t = 2.066$, $p = 0.0404$ **

$N_{\text{pairs}} = 163$

$N_{\text{pairs}} = 163$

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

eq3: gdp_growth

eq3: gdp_growth

$\rho = -0.3230$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$\rho = -0.3230$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$t = 2.328$, $p = 0.0211$ **

$t = 2.328$, $p = 0.0211$ **

$N_{\text{pairs}} = 163$

$N_{\text{pairs}} = 163$

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

→ ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test

TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test

Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
 Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
 Dependence in Panels, Cambridge Working Papers in Economics 0435.
 Dependence in Panels, Cambridge Working Papers in Economics 0435.

$$CD = \sqrt{2T \sum_{i,j} (N(N-1))} \cdot \sum_{i,j} T_{ij} \cdot \rho_{ij}$$

$$CD = \sqrt{2T \sum_{i,j} (N(N-1))} \cdot \sum_{i,j} T_{ij} \cdot \rho_{ij}$$

where ρ_{ij} is the sample correlation of residuals between units i and j ,

where ρ_{ij} is the sample correlation of residuals between units i and j ,

T_{ij} is the number of common time periods.

T_{ij} is the number of common time periods.

Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.

Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.

Note: with $N=8$, the asymptotic distribution is approximate.

Note: with $N=8$, the asymptotic distribution is approximate.

eq1: inst_avg

eq1: inst_avg

CD = -66.7215, $p = 0.0000$ ***

CD = -66.7215, $p = 0.0000$ ***

N pairs = 28, $T = 23$

N pairs = 28, $T = 23$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

CD = -64.7704, $p = 0.0000$ ***

CD = -64.7704, $p = 0.0000$ ***

N pairs = 28, $T = 24$

N pairs = 28, $T = 24$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq3: gdp_growth

eq3: gdp_growth

CD = -72.1120, $p = 0.0000$ ***

CD = -72.1120, $p = 0.0000$ ***

N pairs = 28, $T = 24$

N pairs = 28, $T = 24$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

Reference: Greene (2000), Econometric Analysis, 4th ed., ch. 11.

Reference: Greene (2000), Econometric Analysis, 4th ed., ch. 11.

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

```

eq1: inst_avg
eq1: inst_avg
W = 38.001,  $\chi^2(7)$ ,  $p = 0.0000$  ***
W = 38.001,  $\chi^2(7)$ ,  $p = 0.0000$  ***
 $\sigma^2_i$  by country:
 $\sigma^2_i$  by country:
COL  $\sigma^2 = 0.0686$  T=22.0
COL  $\sigma^2 = 0.0686$  T=22.0
CRI  $\sigma^2 = 0.0196$  T=22.0
CRI  $\sigma^2 = 0.0196$  T=22.0
DOM  $\sigma^2 = 0.0368$  T=21.0
DOM  $\sigma^2 = 0.0368$  T=21.0
GTM  $\sigma^2 = 0.0096$  T=20.0
GTM  $\sigma^2 = 0.0096$  T=20.0
HND  $\sigma^2 = 0.0109$  T=22.0
HND  $\sigma^2 = 0.0109$  T=22.0
NIC  $\sigma^2 = 0.0598$  T=19.0
NIC  $\sigma^2 = 0.0598$  T=19.0
PAN  $\sigma^2 = 0.0133$  T=22.0
PAN  $\sigma^2 = 0.0133$  T=22.0
SLV  $\sigma^2 = 0.0362$  T=23.0
SLV  $\sigma^2 = 0.0362$  T=23.0
Pooled  $\sigma^2 = 0.0316$ 
Pooled  $\sigma^2 = 0.0316$ 
→ ■ Heteroscedastic → clustered / DK SE already applied
→ ■ Heteroscedastic → clustered / DK SE already applied

```

```

eq2: fdi_percent_gdp
eq2: fdi_percent_gdp
W = 67.638,  $\chi^2(7)$ ,  $p = 0.0000$  ***
W = 67.638,  $\chi^2(7)$ ,  $p = 0.0000$  ***
 $\sigma^2_i$  by country:
 $\sigma^2_i$  by country:
COL  $\sigma^2 = 1.0864$  T=23.0
COL  $\sigma^2 = 1.0864$  T=23.0
CRI  $\sigma^2 = 0.5696$  T=23.0
CRI  $\sigma^2 = 0.5696$  T=23.0
DOM  $\sigma^2 = 0.8312$  T=22.0
DOM  $\sigma^2 = 0.8312$  T=22.0
GTM  $\sigma^2 = 1.5648$  T=21.0
GTM  $\sigma^2 = 1.5648$  T=21.0
HND  $\sigma^2 = 0.9044$  T=23.0
HND  $\sigma^2 = 0.9044$  T=23.0
NIC  $\sigma^2 = 1.7567$  T=20.0
NIC  $\sigma^2 = 1.7567$  T=20.0
PAN  $\sigma^2 = 6.1034$  T=23.0
PAN  $\sigma^2 = 6.1034$  T=23.0
SLV  $\sigma^2 = 3.0253$  T=24.0
SLV  $\sigma^2 = 3.0253$  T=24.0
Pooled  $\sigma^2 = 2.0009$ 
Pooled  $\sigma^2 = 2.0009$ 
→ ■ Heteroscedastic → clustered / DK SE already applied
→ ■ Heteroscedastic → clustered / DK SE already applied

eq3: gdp_growth
eq3: gdp_growth

```


$W = 40.945$, $\chi^2(7)$, $p = 0.0000$ ***

$W = 40.945$, $\chi^2(7)$, $p = 0.0000$ ***

σ^2_i by country:

σ^2_i by country:

COL $\sigma^2 = 1.6686$ $T = 23.0$

COL $\sigma^2 = 1.6686$ $T = 23.0$

CRI $\sigma^2 = 1.6430$ $T = 23.0$

CRI $\sigma^2 = 1.6430$ $T = 23.0$

DOM $\sigma^2 = 4.3702$ $T = 22.0$

DOM $\sigma^2 = 4.3702$ $T = 22.0$

GTM $\sigma^2 = 3.4912$ $T = 21.0$

GTM $\sigma^2 = 3.4912$ $T = 21.0$

HND $\sigma^2 = 1.9061$ $T = 23.0$

HND $\sigma^2 = 1.9061$ $T = 23.0$

NIC $\sigma^2 = 3.9101$ $T = 20.0$

NIC $\sigma^2 = 3.9101$ $T = 20.0$

PAN $\sigma^2 = 8.1270$ $T = 23.0$

PAN $\sigma^2 = 8.1270$ $T = 23.0$

SLV $\sigma^2 = 1.0503$ $T = 24.0$

SLV $\sigma^2 = 1.0503$ $T = 24.0$

Pooled $\sigma^2 = 3.2391$

Pooled $\sigma^2 = 3.2391$

→ ■ Heteroscedastic → clustered / DK SE already applied

→ ■ Heteroscedastic → clustered / DK SE already applied

TEST 4 — Multicollinearity: VIF and Condition Number

TEST 4 — Multicollinearity: VIF and Condition Number

VIF computed on within-transformed (two-way demeaned) regressors.

VIF computed on within-transformed (two-way demeaned) regressors.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

eq1: inst_avg

eq1: inst_avg

Variable VIF Flag

Variable VIF Flag

exports_percent_gdp 1.61 ✓ OK

exports_percent_gdp 1.61 ✓ OK

unemployment 1.48 ✓ OK

unemployment 1.48 ✓ OK

population_log 1.43 ✓ OK

population_log 1.43 ✓ OK

homicide_rate_log_lag1 1.34 ✓ OK

homicide_rate_log_lag1 1.34 ✓ OK

gdp_per_capita_log 1.30 ✓ OK

gdp_per_capita_log 1.30 ✓ OK

inflation 1.20 ✓ OK

inflation 1.20 ✓ OK

Condition number: 2.4 ✓ OK (<30)

Condition number: 2.4 ✓ OK (<30)

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

Variable VIF Flag

Variable VIF Flag

exports_percent_gdp 1.68 ✓ OK

exports_percent_gdp 1.68 ✓ OK

population_log 1.42 ✓ OK

population_log 1.42 ✓ OK

gdp_per_capita_log 1.33 ✓ OK

gdp_per_capita_log 1.33 ✓ OK

unemployment 1.22 ✓ OK

unemployment 1.22 ✓ OK

inst_avg 1.19 ✓ OK

inst_avg 1.19 ✓ OK

inflation 1.18 ✓ OK

inflation 1.18 ✓ OK

Condition number: 2.3 ✓ OK (<30)

Condition number: 2.3 ✓ OK (<30)

eq3: gdp_growth

eq3: *gdp_growth*

Variable VIF Flag

Variable VIF Flag

exports_percent_gdp 1.89 ✓ OK

exports_percent_gdp 1.89 ✓ OK

population_log 1.49 ✓ OK

population_log 1.49 ✓ OK

gdp_per_capita_log 1.38 ✓ OK

gdp_per_capita_log 1.38 ✓ OK

fdi_percent_gdp 1.32 ✓ OK

fdi_percent_gdp 1.32 ✓ OK

unemployment 1.26 ✓ OK

unemployment 1.26 ✓ OK

inst_avg 1.21 ✓ OK

inst_avg 1.21 ✓ OK

inflation 1.20 ✓ OK

inflation 1.20 ✓ OK

Condition number: 2.6 ✓ OK (<30)

Condition number: 2.6 ✓ OK (<30)

TEST 5 — Influence Diagnostics: DFBETA by Country

TEST 5 — Influence Diagnostics: DFBETA by Country

$$DFBETA_c = (\beta_{\text{c}} - \beta_{\text{c}}) / SE(\beta_{\text{c}})$$

$$DFBETA_c = (\beta_{\text{c}} - \beta_{\text{c}}) / SE(\beta_{\text{c}})$$

where β_{-c} is the coefficient estimated excluding country c .

where β_{-c} is the coefficient estimated excluding country c .

$|DFBETA| \geq 1$ indicates influential observation.

|DFBETA| > 1 indicates influential observation.
 Reference: Belsley, Kuh & Welsch (1980).
 Reference: Belsley, Kuh & Welsch (1980).

eq1: inst_avg | Key var: homicide_rate_log_lag1

eq1: inst_avg | Key var: homicide_rate_log_lag1

$\beta_{full} = -0.1404$ SE = 0.1358

$\beta_{full} = -0.1404$ SE = 0.1358

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL -0.0197 -0.8887 ✓ OK

COL -0.0197 -0.8887 ✓ OK

CRI -0.2547 0.8411 ✓ OK

CRI -0.2547 0.8411 ✓ OK

DOM -0.0990 -0.3050 ✓ OK

DOM -0.0990 -0.3050 ✓ OK

GTM -0.1523 0.0878 ✓ OK

GTM -0.1523 0.0878 ✓ OK

HND -0.1554 0.1103 ✓ OK

HND -0.1554 0.1103 ✓ OK

NIC -0.1277 -0.0938 ✓ OK

NIC -0.1277 -0.0938 ✓ OK

PAN -0.1087 -0.2334 ✓ OK

PAN -0.1087 -0.2334 ✓ OK

SLV -0.2668 0.9301 ✓ OK

SLV -0.2668 0.9301 ✓ OK

eq2: fdi_percent_gdp | Key var: inst_avg

eq2: fdi_percent_gdp | Key var: inst_avg

$\beta_{full} = 0.9000$ SE = 0.4648

$\beta_{full} = 0.9000$ SE = 0.4648

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL 1.7060 -1.7340 ✗ Influential (|DFBETA|>1)

COL 1.7060 -1.7340 ✗ Influential (|DFBETA|>1)

CRI 0.7640 0.2926 ✓ OK

CRI 0.7640 0.2926 ✓ OK

DOM 0.6559 0.5252 ✓ OK

DOM 0.6559 0.5252 ✓ OK

GTM 0.8322 0.1459 ✓ OK

GTM 0.8322 0.1459 ✓ OK

HND 0.7092 0.4105 ✓ OK

HND 0.7092 0.4105 ✓ OK

NIC 0.7702 0.2792 ✓ OK

NIC 0.7702 0.2792 ✓ OK

PAN 0.8802 0.0426 ✓ OK

PAN 0.8802 0.0426 ✓ OK

SLV 0.6992 0.4321 ✓ OK

SLV 0.6992 0.4321 ✓ OK

eq3: gdp_growth | Key var: fdi_percent_gdp

eq3: gdp_growth | Key var: fdi_percent_gdp

$\beta_{full} = 0.1371$ SE = 0.1095

$\beta_{full} = 0.1371$ SE = 0.1095

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL 0.1020 0.3206 ✓ OK

COL 0.1020 0.3206 ✓ OK

CRI 0.1376 -0.0047 ✓ OK

CRI 0.1376 -0.0047 ✓ OK

DOM 0.2015 -0.5881 ✓ OK

DOM 0.2015 -0.5881 ✓ OK

GTM 0.1572 -0.1838 ✓ OK

GTM 0.1572 -0.1838 ✓ OK

HND 0.1325 0.0418 ✓ OK

HND 0.1325 0.0418 ✓ OK

NIC 0.0754 0.5631 ✓ OK

NIC 0.0754 0.5631 ✓ OK

PAN 0.0077 1.1818 x Influential (|DFBETA|>1)

PAN 0.0077 1.1818 x Influential (|DFBETA|>1)

SLV 0.2082 -0.6493 ✓ OK

SLV 0.2082 -0.6493 ✓ OK

03-F — DIAGNOSTIC FIGURE

03-F — DIAGNOSTIC FIGURE

✓ Figure saved → figures/05_diagnostics.png

✓ Figure saved → figures/05_diagnostics.png

03-G — REMEDIES SUMMARY

03-G — REMEDIES SUMMARY

Diagnostic Result Remedy applied

Diagnostic Result Remedy applied

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Wild bootstrap (Mod.04) ✓

Wild bootstrap (Mod.04) ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Group heterosced. See results above Clustered SE ✓

Group heterosced. See results above Clustered SE ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

Module 03 complete.

Module 03 complete.

✓ Module 03 completed in 11.1s

=====

RUNNING MODULE 04: 04_bootstrap_inference.py

=====

04-A — LOAD

04-A — LOAD

✓ Data loaded. G=8 clusters, B=999 bootstrap draws.

✓ Data loaded. G=8 clusters, B=999 bootstrap draws.

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

■ With G=8, conventional clustered SE over-reject. Wild bootstrap preferred.

■ With G=8, conventional clustered SE over-reject. Wild bootstrap preferred.

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

■■ EQ1: dep=inst_avg, key=homicide_rate_log_lag1 ■■

■■ EQ1: dep=inst_avg, key=homicide_rate_log_lag1 ■■

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

Restricted model: PanelOLS without 'homicide_rate_log_lag1' (H0: $\beta=0$)

Restricted model: PanelOLS without 'homicide_rate_log_lag1' (H0: $\beta=0$)

b=200/999 elapsed=17s valid=200 fails=0

b=200/999 elapsed=17s valid=200 fails=0

b=400/999 elapsed=32s valid=400 fails=0

b=400/999 elapsed=32s valid=400 fails=0

b=600/999 elapsed=51s valid=600 fails=0

b=600/999 elapsed=51s valid=600 fails=0

b=800/999 elapsed=69s valid=800 fails=0

b=800/999 elapsed=69s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

β ■ = -0.1404

β ■ = -0.1404

SE (clustered, conventional) = 0.1358

SE (clustered, conventional) = 0.1358

t = -1.0336

t = -1.0336

Wild cluster bootstrap (Webb, B=999):

Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.4294

p-value (two-sided) = 0.4294

CI 95% (percentile) = [-0.2041, 0.2068]
 CI 95% (percentile) = [-0.2041, 0.2068]
 Valid draws: 999/999 | Fails: 0
 Valid draws: 999/999 | Fails: 0
 Runtime: 86.8s
 Runtime: 86.8s
 ■ $\beta(\text{homicide_rate_log_lag1})$ not significant (bootstrap $p=0.4294$)
 ■ $\beta(\text{homicide_rate_log_lag1})$ not significant (bootstrap $p=0.4294$)
 Note: inspect sign and economic magnitude regardless of p -value.
 Note: inspect sign and economic magnitude regardless of p -value.
 CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 $\beta_{\blacksquare} = -0.1404$
 $\beta_{\blacksquare} = -0.1404$
 $SE(CR2) = 0.1420$
 $SE(CR2) = 0.1420$
 $t(CR2) = -0.9888$
 $t(CR2) = -0.9888$

$df(\text{Satterthwaite}) = 3.16$ [cf. $G-1=7$ used in v1 — different]
 $df(\text{Satterthwaite}) = 3.16$ [cf. $G-1=7$ used in v1 — different]
 $p(CR2) = 0.3922$
 $p(CR2) = 0.3922$
 ■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
 ■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 Restricted model: PanelOLS without 'inst_avg' ($H0: \beta=0$)
 Restricted model: PanelOLS without 'inst_avg' ($H0: \beta=0$)
 b=200/999 elapsed=17s valid=200 fails=0
 b=200/999 elapsed=17s valid=200 fails=0
 b=400/999 elapsed=34s valid=400 fails=0
 b=400/999 elapsed=34s valid=400 fails=0
 b=600/999 elapsed=51s valid=600 fails=0
 b=600/999 elapsed=51s valid=600 fails=0
 b=800/999 elapsed=68s valid=800 fails=0
 b=800/999 elapsed=68s valid=800 fails=0
 Full-sample estimate:
 Full-sample estimate:
 $\beta_{\blacksquare} = 0.9000$
 $\beta_{\blacksquare} = 0.9000$
 $SE(\text{clustered, conventional}) = 0.4648$
 $SE(\text{clustered, conventional}) = 0.4648$
 $t = 1.9363$
 $t = 1.9363$
 Wild cluster bootstrap (Webb, B=999):
 Wild cluster bootstrap (Webb, B=999):
 $p\text{-value (two-sided)} = 0.0621 *$
 $p\text{-value (two-sided)} = 0.0621 *$
 CI 95% (percentile) = [-0.8424, 0.8392]
 CI 95% (percentile) = [-0.8424, 0.8392]
 Valid draws: 999/999 | Fails: 0
 Valid draws: 999/999 | Fails: 0
 Runtime: 85.4s

Runtime: 85.4s

✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap $p=0.0621$)

✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap $p=0.0621$)

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta = 0.9000$

$\beta = 0.9000$

SE(CR2) = 0.5155

SE(CR2) = 0.5155

$t(\text{CR2}) = 1.7460$

$t(\text{CR2}) = 1.7460$

$df(\text{Satterthwaite}) = 2.38$ [cf. G-1=7 used in v1 — different]

$df(\text{Satterthwaite}) = 2.38$ [cf. G-1=7 used in v1 — different]

$p(\text{CR2}) = 0.2023$

$p(\text{CR2}) = 0.2023$

■■ EQ3: dep=gdp_growth, key=fdi_percent_gdp ■■

■■ EQ3: dep=gdp_growth, key=fdi_percent_gdp ■■

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

Restricted model: PanelOLS without 'fdi_percent_gdp' ($H_0: \beta=0$)

Restricted model: PanelOLS without 'fdi_percent_gdp' ($H_0: \beta=0$)

b=200/999 elapsed=19s valid=200 fails=0

b=200/999 elapsed=19s valid=200 fails=0

b=400/999 elapsed=36s valid=400 fails=0

b=400/999 elapsed=36s valid=400 fails=0

b=600/999 elapsed=54s valid=600 fails=0

b=600/999 elapsed=54s valid=600 fails=0

b=800/999 elapsed=71s valid=800 fails=0

b=800/999 elapsed=71s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

$\beta = 0.1371$

$\beta = 0.1371$

SE (clustered, conventional) = 0.1095

SE (clustered, conventional) = 0.1095

$t = 1.2522$

$t = 1.2522$

Wild cluster bootstrap (Webb, B=999):

Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.4094

p-value (two-sided) = 0.4094

CI 95% (percentile) = [-0.2404, 0.2412]

CI 95% (percentile) = [-0.2404, 0.2412]

Valid draws: 999/999 | Fails: 0

Valid draws: 999/999 | Fails: 0

Runtime: 87.6s

Runtime: 87.6s

■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap $p=0.4094$)

■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap $p=0.4094$)

Note: inspect sign and economic magnitude regardless of p-value.

Note: inspect sign and economic magnitude regardless of p-value.

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

```

β■ = 0.1371
β■ = 0.1371
SE(CR2) = 0.1133
SE(CR2) = 0.1133
t(CR2) = 1.2095
t(CR2) = 1.2095
df(Satterthwaite) = 1.79 [cf. G-1=7 used in v1 — different]
df(Satterthwaite) = 1.79 [cf. G-1=7 used in v1 — different]
p(CR2) = 0.3618
p(CR2) = 0.3618

```

04-C — SE COMPARISON TABLE

04-C — SE COMPARISON TABLE

```

Equation Dep. var. Key var. β■ SE(Cl.) p(Cl.) SE(DK) SE(CR2) df_satt p(WCB)
Equation Dep. var. Key var. β■ SE(Cl.) p(Cl.) SE(DK) SE(CR2) df_satt p(WCB)

```

```

eq1 inst_avg homicide_rate_log_lag1 -0.1404 0.1358 0.303 0.1282 0.1420 3.16 0.429
eq1 inst_avg homicide_rate_log_lag1 -0.1404 0.1358 0.303 0.1282 0.1420 3.16 0.429
eq2 fdi_percent_gdp inst_avg 0.9000 0.4648 0.055 0.5690 0.5155 2.38 0.062*
eq2 fdi_percent_gdp inst_avg 0.9000 0.4648 0.055 0.5690 0.5155 2.38 0.062*
eq3 gdp_growth fdi_percent_gdp 0.1371 0.1095 nan nan 0.1133 1.79 0.409
eq3 gdp_growth fdi_percent_gdp 0.1371 0.1095 nan nan 0.1133 1.79 0.409

```

Columns:

Columns:

SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

SE(CR2) with df_Satterthwaite as secondary check.

SE(CR2) with df_Satterthwaite as secondary check.

04-D — BOOTSTRAP FIGURES

04-D — BOOTSTRAP FIGURES

```

✓ Figure saved → figures/06_bootstrap_distributions.png
✓ Figure saved → figures/06_bootstrap_distributions.png
✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_bootstrap.json

```


✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_bootstrap.json

Module 04 complete.

Module 04 complete.

✓ Module 04 completed in 267.9s

=====

RUNNING MODULE 05: 05_robustness.py

=====

05-A — LOAD

05-A — LOAD

✓ Data: 179 obs | 8 countries

✓ Data: 179 obs | 8 countries

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

Bias addressed: results driven by a single influential country.

Bias addressed: results driven by a single influential country.

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample (baseline) 171 -0.1404 0.1358 0.303 +0.0000

Full sample (baseline) 171 -0.1404 0.1358 0.303 +0.0000

Excl. COL 149 -0.0197 0.1164 0.866 +0.1207

Excl. COL 149 -0.0197 0.1164 0.866 +0.1207

Excl. CRI 149 -0.2547 0.1383 0.068 * -0.1143

Excl. CRI 149 -0.2547 0.1383 0.068 * -0.1143

Excl. DOM 150 -0.0990 0.1569 0.529 +0.0414

Excl. DOM 150 -0.0990 0.1569 0.529 +0.0414

Excl. GTM 151 -0.1523 0.1396 0.277 -0.0119

Excl. GTM 151 -0.1523 0.1396 0.277 -0.0119

Excl. HND 149 -0.1554 0.1490 0.299 -0.0150

Excl. HND 149 -0.1554 0.1490 0.299 -0.0150

Excl. NIC 152 -0.1277 0.0930 0.172 +0.0127

Excl. NIC 152 -0.1277 0.0930 0.172 +0.0127

Excl. PAN 149 -0.1087 0.1401 0.439 +0.0317

Excl. PAN 149 -0.1087 0.1401 0.439 +0.0317

Excl. SLV 148 -0.2668 0.1241 0.034 ** -0.1264

Excl. SLV 148 -0.2668 0.1241 0.034 ** -0.1264

■ 0 sign reversal(s), 3 large shift(s) — check influential countries.

■ 0 sign reversal(s), 3 large shift(s) — check influential countries.

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

<https://www.pearsoncmg.com/api/v1/print/healthcare/9780134160199>

ROBUSTNESS 2 — Time Windows

ROBUSTNESS 2 — Time Windows

Bias addressed: period-specific shocks, structural breaks.

Bias addressed: period-specific shocks, structural breaks.

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample 171 -0.1404 0.1358 0.303 +0.0000

Full sample 171 -0.1404 0.1358 0.303 +0.0000

Pre-2010 (pre-GFC) 64 -0.1277 0.1443 0.381 +0.0127

Pre-2010 (pre-GFC) 64 -0.1277 0.1443 0.381 +0.0127

Post-2010 107 0.0252 0.0660 0.704 +0.1656

Post-2010 107 0.0252 0.0660 0.704 +0.1656

Excl. COVID (2020–21) 156 -0.1042 0.1188 0.382 +0.0362

Excl. COVID (2020–21) 156 -0.1042 0.1188 0.382 +0.0362

Post-peace (2014+) 75 0.0797 0.1053 0.453 +0.2201

Post-peace (2014+) 75 0.0797 0.1053 0.453 +0.2201

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample 179 0.9000 0.4648 0.055 * +0.0000

Full sample 179 0.9000 0.4648 0.055 * +0.0000

Pre-2010 (pre-GFC) 72 -0.0124 0.8363 0.988 -0.9124

Pre-2010 (pre-GFC) 72 -0.0124 0.8363 0.988 -0.9124

Post-2010 107 1.9317 0.9407 0.043 ** +1.0317

Post-2010 107 1.9317 0.9407 0.043 ** +1.0317

Excl. COVID (2020–21) 164 1.0179 0.3159 0.002 *** +0.1179

Excl. COVID (2020–21) 164 1.0179 0.3159 0.002 *** +0.1179

Post-peace (2014+) 75 3.6099 0.7574 0.000 *** +2.7099

Post-peace (2014+) 75 3.6099 0.7574 0.000 *** +2.7099

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample 179 0.1371 0.1095 0.213 +0.0000

Full sample 179 0.1371 0.1095 0.213 +0.0000

Pre-2010 (pre-GFC) 72 0.0512 0.1244 0.682 -0.0859

Pre-2010 (pre-GFC) 72 0.0512 0.1244 0.682 -0.0859
 Post-2010 107 0.2680 0.1771 0.134 +0.1309
 Post-2010 107 0.2680 0.1771 0.134 +0.1309
 Excl. COVID (2020–21) 164 -0.0079 0.0945 0.933 -0.1450
 Excl. COVID (2020–21) 164 -0.0079 0.0945 0.933 -0.1450
 Post-peace (2014+) 75 0.0842 0.2595 0.747 -0.0529
 Post-peace (2014+) 75 0.0842 0.2595 0.747 -0.0529

ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)
 ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)

Uses pre-computed lag columns from dataset (not re-computed).
 Uses pre-computed lag columns from dataset (not re-computed).
 Bias addressed: uncertainty about transmission delay of violence.
 Bias addressed: uncertainty about transmission delay of violence.

EQ1: Alternative Lags

EQ1: Alternative Lags

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Contemporaneous (lag 0) 179 -0.0512 0.1080 0.636 +0.0892
 Contemporaneous (lag 0) 179 -0.0512 0.1080 0.636 +0.0892
 Lag 1 (primary) 171 -0.1404 0.1358 0.303 +0.0000
 Lag 1 (primary) 171 -0.1404 0.1358 0.303 +0.0000
 Lag 2 163 -0.1930 0.1670 0.250 -0.0526
 Lag 2 163 -0.1930 0.1670 0.250 -0.0526
 Lag 3 155 -0.1836 0.1813 0.313 -0.0432
 Lag 3 155 -0.1836 0.1813 0.313 -0.0432

ROBUSTNESS 4 — Alternative Institution Indices

ROBUSTNESS 4 — Alternative Institution Indices

Bias addressed: sensitivity to index construction methodology.
 Bias addressed: sensitivity to index construction methodology.

EQ1: Alternative Institution Indices

EQ1: Alternative Institution Indices

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

inst_avg (primary) 171 -0.1404 0.1358 0.303 +0.0000
 inst_avg (primary) 171 -0.1404 0.1358 0.303 +0.0000
 inst_pca (PCA-based) 171 -0.2333 0.2264 0.305 -0.0929
 inst_pca (PCA-based) 171 -0.2333 0.2264 0.305 -0.0929

rule_of_law 171 -1.8345 1.5064 0.225 -1.6941
 rule_of_law 171 -1.8345 1.5064 0.225 -1.6941
 control_corruption 171 0.8270 0.9159 0.368 +0.9674
 control_corruption 171 0.8270 0.9159 0.368 +0.9674
 political_stability 171 -3.3434 3.1976 0.298 -3.2030
 political_stability 171 -3.3434 3.1976 0.298 -3.2030
 ■ Mixed signs across institution specifications — investigate.
 ■ Mixed signs across institution specifications — investigate.

ROBUSTNESS 5 — Alternative Control Sets
 ROBUSTNESS 5 — Alternative Control Sets

Bias addressed: over-controlling or omitted variable bias.
 Bias addressed: over-controlling or omitted variable bias.

EQ1: inst_avg

EQ1: inst_avg

Specification N ■ SE p Stars Δβ

Specification N ■ SE p Stars Δβ

Baseline controls 171 -0.1404 0.1358 0.303 +0.0000

Baseline controls 171 -0.1404 0.1358 0.303 +0.0000

Extended controls — — — — FAILED

Extended controls — — — — FAILED

Minimal controls 171 -0.1034 0.1475 0.484 +0.0370

Minimal controls 171 -0.1034 0.1475 0.484 +0.0370

No controls (FE only) 171 -0.0825 0.1702 0.629 +0.0579

No controls (FE only) 171 -0.0825 0.1702 0.629 +0.0579

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N ■ SE p Stars Δβ

Specification N ■ SE p Stars Δβ

Baseline controls 179 0.9000 0.4648 0.055 * +0.0000

Baseline controls 179 0.9000 0.4648 0.055 * +0.0000

Extended controls — — — — FAILED

Extended controls — — — — FAILED

Minimal controls 179 0.5491 0.8413 0.515 -0.3509

Minimal controls 179 0.5491 0.8413 0.515 -0.3509

No controls (FE only) 179 -0.1192 0.8270 0.886 -1.0192

No controls (FE only) 179 -0.1192 0.8270 0.886 -1.0192

EQ3: gdp_growth

EQ3: gdp_growth

Specification N ■ SE p Stars Δβ

Specification N ■ SE p Stars Δβ

```

Baseline controls 179 0.1450 0.1212 0.234 +0.0000
Baseline controls 179 0.1450 0.1212 0.234 +0.0000
Extended controls — — — — FAILED
Extended controls — — — — FAILED
Minimal controls 179 0.2666 0.1352 0.050 * +0.1216
Minimal controls 179 0.2666 0.1352 0.050 * +0.1216
No controls (FE only) 179 0.2355 0.1354 0.084 * +0.0905
No controls (FE only) 179 0.2355 0.1354 0.084 * +0.0905

```

```

ROBUSTNESS 6 — Placebo Test (permuted homicide)
ROBUSTNESS 6 — Placebo Test (permuted homicide)

```

```

Procedure: randomly permute homicide_rate_log_lag1 within year cells.
Procedure: randomly permute homicide_rate_log_lag1 within year cells.
This breaks the country-specific causal link while preserving the
This breaks the country-specific causal link while preserving the
cross-sectional and temporal distribution of the variable.
cross-sectional and temporal distribution of the variable.
H0 (placebo): permuted violence has no effect on institutions.
H0 (placebo): permuted violence has no effect on institutions.
If the TRUE estimate is more extreme than 95% of placebo estimates,
If the TRUE estimate is more extreme than 95% of placebo estimates,
this confirms the result is not driven by spurious correlation.
this confirms the result is not driven by spurious correlation.
N_permutations = 500.
N_permutations = 500.

True estimate:  $\beta_{\text{H}} = -0.1404$ 
True estimate:  $\beta_{\text{H}} = -0.1404$ 
Placebo distribution: mean=0.0006, SD=0.0243, 5th pctl=-0.0404
Placebo distribution: mean=0.0006, SD=0.0243, 5th pctl=-0.0404
Placebo p-value: 0.0000
Placebo p-value: 0.0000
✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.
✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

```

```

05-G — FIGURES
05-G — FIGURES

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✓ Figure saved → figures/07_robustness.png

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05-H — EXPORT
05-H — EXPORT

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✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\05_robustness.json

Module 05 complete.

Module 05 complete.

✓ Module 05 completed in 60.2s

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RUNNING MODULE 06: 06_ml_triangular.py

=====

06-A — LOAD

06-A — LOAD

✓ Data: 179 obs | 8 countries

✓ Data: 179 obs | 8 countries

CV strategy: Leave-One-Country-Out (8 folds)

CV strategy: Leave-One-Country-Out (8 folds)

■ All ML results are EXPLORATORY. They do not imply causation.

■ All ML results are EXPLORATORY. They do not imply causation.

■ G=8 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

■ G=8 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

■■ T1: Violence → Institutions ■■

■■ T1: Violence → Institutions ■■

Missing summary:

Missing summary:

homicide_rate_log_lag2 0.089385

homicide_rate_log_lag2 0.089385

homicide_rate_log_lag1 0.044693

homicide_rate_log_lag1 0.044693

inst_avg 0.000000

inst_avg 0.000000

gdp_per_capita_log 0.000000

gdp_per_capita_log 0.000000

inflation 0.000000

inflation 0.000000

unemployment 0.000000

unemployment 0.000000

exports_percent_gdp 0.000000

exports_percent_gdp 0.000000

population_log 0.000000

population_log 0.000000

year_c 0.000000

year_c 0.000000

dtype: float64

dtype: float64

Rows before dropna: 179

Rows before dropna: 179

Rows after dropna: 163

Rows after dropna: 163

Model R^2 mean R^2 SD RMSE mean Valid folds

Model R^2 mean R^2 SD RMSE mean Valid folds

Random Forest -33.1112 49.1064 0.7235 8

Random Forest -33.1112 49.1064 0.7235 8

Gradient Boosting -49.0544 74.8629 0.8054 8

Gradient Boosting -49.0544 74.8629 0.8054 8

■ Note: G=8 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

■ Note: G=8 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

N pairs: 8 | W=16.0 p=0.84375

N pairs: 8 | W=16.0 p=0.84375

■ G=8 pairs → very low power. Treat p-value as descriptive only.

■ G=8 pairs → very low power. Treat p-value as descriptive only.

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.8181

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.8181

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Population (log) 0.3159 0.3473±0.0335 0.2440 1.61 0.93 0.96 0.99

Population (log) 0.3159 0.3473±0.0335 0.2440 1.61 0.93 0.96 0.99

GDP per capita (log) 0.2029 0.1752±0.0227 0.1676 2.52 1.42 0.73 0.94

GDP per capita (log) 0.2029 0.1752±0.0227 0.1676 2.52 1.42 0.73 0.94

Violence lag 1 0.1314 0.0721±0.0106 0.1077 4.34 1.26 0.23 0.84

Violence lag 1 0.1314 0.0721±0.0106 0.1077 4.34 1.26 0.23 0.84

Violence lag 2 0.1264 0.0703±0.0097 0.1020 3.17 1.52 0.70 0.90

Violence lag 2 0.1264 0.0703±0.0097 0.1020 3.17 1.52 0.70 0.90

Unemployment 0.0789 0.0576±0.0093 0.0650 5.34 1.43 0.10 0.57

Unemployment 0.0789 0.0576±0.0093 0.0650 5.34 1.43 0.10 0.57

Exports (% GDP) 0.0780 0.0492±0.0045 0.0429 5.03 1.55 0.17 0.51

Exports (% GDP) 0.0780 0.0492±0.0045 0.0429 5.03 1.55 0.17 0.51

Inflation 0.0476 0.0323±0.0041 0.0341 6.28 1.38 0.10 0.23

Inflation 0.0476 0.0323±0.0041 0.0341 6.28 1.38 0.10 0.23

Time Trend (centred) 0.0188 0.0119±0.0019 0.0188 7.71 0.65 0.00 0.02

Time Trend (centred) 0.0188 0.0119±0.0019 0.0188 7.71 0.65 0.00 0.02

Key feature 'homicide_rate_log_lag1':

Key feature 'homicide_rate_log_lag1':

Permutation rank: #3/8

Permutation rank: #3/8

SHAP rank: #3/8

SHAP rank: #3/8

✓ Key feature ranks in top 3 by both importance metrics.

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)
 Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

```

Exports (% GDP) 0.3082 0.3099±0.0374 0.7221 1.10 0.50 0.99 0.99
Exports (% GDP) 0.3082 0.3099±0.0374 0.7221 1.10 0.50 0.99 0.99
Violence lag 1 0.1420 0.0962±0.0162 0.3994 3.14 1.51 0.67 0.92
Violence lag 1 0.1420 0.0962±0.0162 0.3994 3.14 1.51 0.67 0.92
Institution Index (avg) 0.1106 0.0741±0.0114 0.2509 4.23 1.77 0.41 0.77
Institution Index (avg) 0.1106 0.0741±0.0114 0.2509 4.23 1.77 0.41 0.77
Unemployment 0.1618 0.1178±0.0163 0.2487 4.20 1.64 0.41 0.76
Unemployment 0.1618 0.1178±0.0163 0.2487 4.20 1.64 0.41 0.76

```

```

Tourist Arrivals (log) 0.0810 0.0624±0.0100 0.1661 5.14 1.13 0.08 0.56
Tourist Arrivals (log) 0.0810 0.0624±0.0100 0.1661 5.14 1.13 0.08 0.56
GDP per capita (log) 0.0905 0.0480±0.0063 0.1578 4.67 1.66 0.28 0.64
GDP per capita (log) 0.0905 0.0480±0.0063 0.1578 4.67 1.66 0.28 0.64
Time Trend (centred) 0.0682 0.0565±0.0092 0.1503 7.45 0.70 0.00 0.01
Time Trend (centred) 0.0682 0.0565±0.0092 0.1503 7.45 0.70 0.00 0.01
Inflation 0.0376 0.0213±0.0037 0.0815 6.08 1.99 0.16 0.35
Inflation 0.0376 0.0213±0.0037 0.0815 6.08 1.99 0.16 0.35

```

Key feature 'inst_avg':

Key feature 'inst_avg':

Permutation rank: #4/8

Permutation rank: #4/8

SHAP rank: #3/8

SHAP rank: #3/8

■ Key feature does not rank in top 3. Inspect collinear features.

■ Key feature does not rank in top 3. Inspect collinear features.

■■ T3: FDI + Institutions → Growth ■■

■■ T3: FDI + Institutions → Growth ■■

Missing summary:

Missing summary:

homicide_rate_log_lag1 0.044693

homicide_rate_log_lag1 0.044693

gdp_growth 0.000000

gdp_growth 0.000000

fdi_percent_gdp 0.000000

fdi_percent_gdp 0.000000

inst_avg 0.000000

inst_avg 0.000000

gdp_per_capita_log 0.000000

gdp_per_capita_log 0.000000

inflation 0.000000

inflation 0.000000

unemployment 0.000000

unemployment 0.000000

exports_percent_gdp 0.000000

exports_percent_gdp 0.000000

year_c 0.000000

year_c 0.000000

dtype: float64

dtype: float64

Rows before dropna: 179

Rows before dropna: 179

Rows after dropna: 171

Rows after dropna: 171

Model R^2 mean R^2 SD RMSE mean Valid folds

Model R^2 mean R^2 SD RMSE mean Valid folds

Random Forest 0.0989 0.1458 3.4406 8

Random Forest 0.0989 0.1458 3.4406 8

Gradient Boosting 0.2418 0.4056 2.9701 8

Gradient Boosting 0.2418 0.4056 2.9701 8

■ Note: G=8 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

■ Note: G=8 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

N pairs: 8 | W=11.0 p=0.3828125

N pairs: 8 | W=11.0 p=0.3828125

■ G=8 pairs → very low power. Treat p-value as descriptive only.

■ G=8 pairs → very low power. Treat p-value as descriptive only.

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.1995

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.1995

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

FDI (% GDP) 0.1833 0.1102±0.0147 0.3284 1.98 1.13 0.91 0.98

FDI (% GDP) 0.1833 0.1102±0.0147 0.3284 1.98 1.13 0.91 0.98

Time Trend (centred) 0.1988 0.1760±0.0195 0.3029 4.17 1.65 0.42 0.80

Time Trend (centred) 0.1988 0.1760±0.0195 0.3029 4.17 1.65 0.42 0.80

Exports (% GDP) 0.1438 0.0743±0.0125 0.3021 3.02 2.00 0.62 0.86

Exports (% GDP) 0.1438 0.0743±0.0125 0.3021 3.02 2.00 0.62 0.86

Inflation 0.1300 0.0677±0.0123 0.2243 4.30 1.89 0.34 0.69

Inflation 0.1300 0.0677±0.0123 0.2243 4.30 1.89 0.34 0.69

GDP per capita (log) 0.0818 0.0407±0.0073 0.2009 5.20 1.70 0.18 0.55

GDP per capita (log) 0.0818 0.0407±0.0073 0.2009 5.20 1.70 0.18 0.55

Unemployment 0.1173 0.0540±0.0074 0.1391 5.58 1.86 0.16 0.44

Unemployment 0.1173 0.0540±0.0074 0.1391 5.58 1.86 0.16 0.44

Violence lag 1 0.0702 0.0315±0.0060 0.1357 6.16 2.19 0.17 0.28

Violence lag 1 0.0702 0.0315±0.0060 0.1357 6.16 2.19 0.17 0.28

Institution Index (avg) 0.0748 0.0260±0.0037 0.0917 5.58 2.23 0.19 0.41

Institution Index (avg) 0.0748 0.0260±0.0037 0.0917 5.58 2.23 0.19 0.41

Key feature 'fdi_percent_gdp':

Key feature 'fdi_percent_gdp':

Permutation rank: #2/8

Permutation rank: #2/8

SHAP rank: #1/8

SHAP rank: #1/8

✓ Key feature ranks in top 3 by both importance metrics.

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06-C — CONVERGENCE TABLE: FE vs ML

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Convergence principle (Athey & Imbens 2019):

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If the variables that matter causally (significant in FE) also rank

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highly in ML importance, this triangulation strengthens the narrative.

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Divergence signals potential non-linearities or confounders.

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Task FE coef FE p-val FE sig Direction Perm rank SHAP rank Convergence

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T1: Violence → Institutions ? 0.3032 3 3 ✓ Strong

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T2: Institutions → FDI ? 0.0548 * 4 3 ■ Partial

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T3: FDI + Institutions → Growth ? ? ? 2 1 ✓ Strong

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Notes:

Notes:

- FE coef and p-value are cluster-robust estimates from Module 02.

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- FE sig: * p<0.10 ** p<0.05 *** p<0.01.

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- Direction: sign of the FE coefficient for the key feature.

- Direction: sign of the FE coefficient for the key feature.

- Perm/SHAP rank: rank among all features in the ML model (1 = most important).

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- Convergence: Strong = key feature ranks top 3 by both metrics.

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06-D — FIGURES

06-D — FIGURES

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/10_shap_ale.png

✓ Figure saved → figures/10_shap_ale.png

06-E — EXPORT

06-E — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\06_ml_results.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\06_ml_results.json

Module 06 complete. Pipeline finished.

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■ IMPORTANT REMINDER ■

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■ ML results (SHAP, permutation importance, LOCO R^2) ■

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■ are EXPLORATORY and NON-CAUSAL. ■

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■ They triangulate — they do not identify causal effects. ■

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■ Primary causal evidence: Module 02 (Two-Way FE) and ■

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■ Module 04 (Wild Cluster Bootstrap inference). ■

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■ ■

■ ■

■ SHAP limitation: computed on the full-sample model. ■

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■ Values reflect in-sample feature associations and may not ■

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■ generalise to held-out countries. ■

■ generalise to held-out countries. ■

■ ■

■ ■

■ Bootstrap stability limitation: $G=8$ clusters → discrete ■

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■ bootstrap distribution. Treat rank SDs and hit rates as ■

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■ approximate lower bounds on true uncertainty. ■

■ approximate lower bounds on true uncertainty. ■

✓ Module 06 completed in 1040.7s

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PIPELINE SUMMARY

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✓ Module 01: Data Preparation & Institution Index

✓ Module 02: Panel Estimation (Two-Way FE)

✓ Module 03: Panel Diagnostics

✓ Module 04: Wild Cluster Bootstrap Inference

✓ Module 05: Robustness Checks

✓ Module 06: ML Triangulation

Total runtime: 1432.6s

Output directories:

Figures → C:\data_analyzer\econometric_pipeline\pipeline\figures\

Tables → C:\data_analyzer\econometric_pipeline\pipeline\tables/

JSON → C:\data_analyzer\econometric_pipeline\pipeline\json/

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